

		FIR VEHICLE TESTING		26/05/2025
FIR No :FIR/Testing-Outdoor-UA/520700/2025/00415				
Part No/ Group :52075256R				
Subject :Report on Fuel Economy evaluation of Signa 3523.T BSVI-II with AC ON vehicle in comparison with Signa 3523.T BSVI-II AC OFF base vehicle.				
Test Vehicle	Project Name :Signa 3523.T BSVI-II with AC ON Model Name : Signa 3523.T BSVI-II			Enclosures : Annexure 1-8
	Chasis Number :PC53562032RJ00873 Engine Number :22L63919785			
FIR Status :		ESO FIR		

Role	Name	Signature
Prepared By	Palash Das	 26-May-2025
Checked By	Swapnil Jadhav	 26-May-2025
Approved By	Ajaykumar Jadhav	 26-May-2025

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

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Documents : Please click here

[Report on Fuel Economy ev..](#)

 FIR/Testing-Outdoor-UA/520700/2025/00415	FIR VEHICLE TESTING		Date: 20/05/2025	
Part No / Group: 52075256R	Specification no :- " 259052070002 "		Action by	Failure class

Subject: -	Report on Fuel Economy evaluation of Signa 3523.T BSVI-II with AC ON vehicle in comparison with Signa 3523.T BSVI-II AC OFF base vehicle.																											
1.0 <u>Conclusion:</u> 1.1 <u>Fuel economy in laden (rated load) at 40 kmph with AC ON condition:</u> - Fuel economy of Signa 3523.T BSVI-II vehicle with AC ON found to be 3.47 kmpl, which is inferior by 3.7% wrt Signa 3523.T BSVI-II AC OFF base Vehicle. 1.2 <u>Fuel economy in Unladen at 40 kmph with AC ON condition:</u> - Fuel economy of Signa 3523.T BSVI-II vehicle with AC ON found to be 6.29 kmpl, which is inferior by 3.7% wrt Signa 3523.T BSVI-II AC OFF base Vehicle. 1.3 <u>Overall fluid economy status at 40 kmph (80% Laden + 20% unladen condition):</u> - Fluid economy of Signa 3523.T BSVI-II vehicle with AC ON found to be 3.81 kmpl, which is inferior by 3.7% wrt Signa 3523.T BSVI-II AC OFF base Vehicle. Note: - - Test vehicle Signa 3523.T BSVI-II vehicle run in AC ON & OFF condition whereas Base Vehicle Signa 3523.T BSVI-II Vehicle run in AC off condition only to determine the actual Delta of AC ON wrt AC OFF. - As Per worse case matrix, Signa 3523.T vehicle is validated & AC ON FE delta (Drop) extended to following vehicle models (Refer Annexure No- 6)		VI/ ENG	C																									
		VI/ ENG	C																									
		VI/ ENG	C																									
<table><tr><th colspan="5">Extension Matrix Models</th></tr><tr><th>SI No</th><th>Vehicle PAT</th><th>Engine</th><th>Lead vehicle/Vehicle Tested</th><th>Variant</th></tr><tr><td>10</td><td rowspan="6">FE</td><td rowspan="6">5.6L (230HP)</td><td rowspan="6">SIGNA/LPT 3523.T</td><td>SIGNA/LPT 3723.T</td></tr><tr><td>11</td><td>SIGNA/LPT 3523.T</td></tr><tr><td>12</td><td>SIGNA/LPT 3123.T</td></tr><tr><td>13</td><td>SIGNA/LPT 3023.T</td></tr><tr><td>14</td><td>SIGNA/LPT 2823.T</td></tr><tr><td>15</td><td>SIGNA/LPT 2623.T</td></tr></table>				Extension Matrix Models					SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	10	FE	5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA/LPT 3723.T	11	SIGNA/LPT 3523.T	12	SIGNA/LPT 3123.T	13	SIGNA/LPT 3023.T	14	SIGNA/LPT 2823.T	15	SIGNA/LPT 2623.T
Extension Matrix Models																												
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant																								
10	FE	5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA/LPT 3723.T																								
11				SIGNA/LPT 3523.T																								
12				SIGNA/LPT 3123.T																								
13				SIGNA/LPT 3023.T																								
14				SIGNA/LPT 2823.T																								
15				SIGNA/LPT 2623.T																								
2.0 <u>Result:</u> Vide Observations																												
FIR Status			Page 1 of 4																									

 FIR/Testing-Outdoor-UA/520700/2025/00415	FIR VEHICLE TESTING		Date: 20/05/2025	
Part No / Group: 52075256R	Specification no :- " 259052070002 "		Action by	Failure class

FAILURES CLASS: A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement. B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (-- First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention. C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule. D: Customer perception issues and minor irritants.											
3.0 Test Vehicle: 1. Signa 3523.T BSVI-II (ERC-1309) Chassis No. – PC53562032RJ00873 Engine No. – 22L63919785		Enclosures: Annexure 1: Panel chart Annexure 2: PAT Sheet Annexure 3: PAT Summary Annexure 4: Test request Annexure 5: Test plan Annexure 6: Mail Comm. for Extension matrix & Cal ID Annexure 7: Cal ID Annexure 8: Compressor DC									
Input Doc Ref.: - Testing-Outdoor-UA/01326 DID No: – NA. DDP No: – NA.											
Work Order No. :- P.24.EA.550	Make :- TATA Motors Limited	T I No. UA001188									
Earlier Ref: - NA											
4.0 Project Status:- DR3											
5.0 Introduction: As a part of AC regulation program, Signa 3523.T BSVI-II vehicle has been offered for AC ON Fuel Economy evaluation with Base Vehicle Signa 3523.T BSVI-II AC OFF base Vehicle. This report provides a comprehensive overview of the trial details and its status.											
6.0 Testing: Following test was conducted during evaluation :											
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 10%;">Sr. no.</th> <th style="width: 40%;">Test</th> <th style="width: 30%;">Reference standard</th> <th style="width: 20%;">Instruments used</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">1</td> <td>Fuel economy evaluation</td> <td style="text-align: center;">TST/VT/WG/158</td> <td style="text-align: center;">Cummins Data Logger</td> </tr> </tbody> </table>	Sr. no.	Test	Reference standard	Instruments used	1	Fuel economy evaluation	TST/VT/WG/158	Cummins Data Logger			
Sr. no.	Test	Reference standard	Instruments used								
1	Fuel economy evaluation	TST/VT/WG/158	Cummins Data Logger								
FIR Status			Page 2 of 4								

 FIR/Testing-Outdoor-UA/520700/2025/00415		FIR VEHICLE TESTING		Date: 20/05/2025		
Part No / Group: 52075256R		Specification no :- " 259052070002 "			Action by	Failure class

7.0 Vehicle details:

Parameter		Test vehicle
Vehicle model		Signa 3523.T BSVI-II (ERC-1309)
Engine Model		Cummins ISBe 5.6L/ Ashwamedh
Calibration ID		PF10040
CAL Used in Vehicle		XWTAA19D07
Max Power		220 HP @2300 RPM
Max Torque		925 Nm @ 1000 ~1300 rpm
GVW (kg)		35000
Gear Box		TATA G-950 DD
Gear Ratios.		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00
Axle Ratio		5.12
Tyres	FA1	Size: : 295/90R 20, Make: M/s Bridgestone, Type : Phase-II
	RA1	Size: : 295/90R 20, Make: M/s Bridgestone, Type : Phase-II
	RA2	Size: : 295/90R 20, Make: M/s Bridgestone, Type : Phase-II
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"

Parameter	Test vehicle
Vehicle model	Signa 3523.T BSVI-II (ERC-1309)
Base Cal Improvement (Cal Maturity %)	100% Cal Maturity
Low Viscous Engine Oil (10W30)	--
FE Switch	NA (iUSFE)
Engine Features (EOL : Enable/Disable)	Enable
ATS Optimization	--
APM	--
Cruise Control	No
Gear based VAM	No
Intelligent mode shifter	No
Moving Mass Estimation	No
Driveline Optimization	Yes
GSA (Gear Shift Advisor)	Yes
MSA (Mode Shift Advisor)	Yes
Low CRR Tyres	Yes
Low Viscous rear axle oil (80W90)	Yes
Smart Alternator (LIN)	No
THU/Unitised/Low friction bearing	No

CALIBRATION ID:

Vehicle PAT	Engine	Lead vehicle / Vehicle Tested	Truck Models	Engine Cal Id
FE	5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA/LPT 3723.T	PF10040
			SIGNA/LPT 3523.T	PF10040
			SIGNA/LPT 3123.T	PF10040
			SIGNA/LPT 3023.T	PF10040
			SIGNA/LPT 2823.T	PF10040/PF10046
			SIGNA/LPT 2623.T	PF10040

FIR Status

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 FIR/Testing-Outdoor-UA/520700/2025/00415		FIR VEHICLE TESTING		Date: 20/05/2025			
Part No / Group: 52075256R		Specification no :- " 259052070002 "			<table><tr><td>Action by</td><td>Failure class</td></tr></table>	Action by	Failure class
Action by	Failure class						

8.0 Test result: Cruising speed at 40 km/hr with AC ON & OFF Condition:													
Sr. No.	Parameters			Base Vehicle : OBD-2					Test Vehicle : OBD-2				
				LPT 3523.T Tata RDE BSVI					SIGNA 3523.T Tata RDE BSVI				
1	Vehicle Identification No. (ERC No./EJ No./Reg. No.)			ERC-1310					ERC-1309				
2	Vehicle GVW/GCW		kg	35000					35000				
3	Engine Details (Make/cc/Norms/no. of Cyl.)			Cummins ISBe 5.6L/ Ashwamedh					Cummins ISBe 5.6L/ Ashwamedh				
	Engine Max Power			220 hp @ 2300 rpm					220 hp @ 2300 rpm				
	Engine Max Torque			925 Nm @ 1000-1300 rpm					925 Nm @ 1000-1300 rpm				
4	Engine Calibration Maturity Level			100% Cal Maturity					100% Cal Maturity				
	Regeneration Strategy (Time ase/Soot base)			Time Based (100 Hrs)					Time Based (100 Hrs)				
5	Gear Box (make/drive/FGR)			TATA G-950 DD					TATA G-950 DD				
6	Gear Ratios			1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00					1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00				
7	RAR			5.12					5.12				
8	Tyre Details			295/90R 20 RIB Bridgestone Low LCRR					295/90R 20 RIB Bridgestone Low LCRR				
	LBSC	High & Low Engine Speed Breakpoint		RPM	1900 rpm, 2000 rpm					1900 rpm, 2000 rpm			
		Vehicle Mass Threshold		kg	35000					35000			
	VAM	Variable Acceleration 1, 2		kmph/sec	3.0,2.5					3.0,2.5			
		Variable Acceleration Speed 1, 2		kmph	10, 40					10, 40			
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)		kmph	50, 50					50,50			
		Maximum Accelerator Vehicle Speed		kmph	80					80			
	RSG	Torque Limit Medium Adjustment Factor		Balance	120%					120%			
		Torque Limit Maximum Adjustment Factor		Economy	130%					130%			
	PTP	Powertrain Protection (Enable / Disable)			Disable					Disable			
		Torque Limit		Nm	NA					NA			
	9	USFE / Torque Table Revision (1/2/3/n)			USFE-1					USFE-1			
		Fan Details (type/make/blades)			24" High Efficiency Viscous fan					24" High Efficiency Viscous fan			
		Base Cal Improvement (Cal Maturity %)			100% Cal Maturity					100% Cal Maturity			
		Low Viscous Engine Oil (TMGO-10W30)			Yes (1SW40CK4+)					Yes (1SW40CK4+)			
FE Switch			NA					NA					
Engine Features (EOL : Enable/Disable)			Enable					Enable					
ATS Optimization			Yes					Yes					
APM			Yes					Yes					
Cruise Control			No					No					
Gear based VAM			No					No					
Intelligent mode shifter			No					No					
Moving Mass Estimation			Yes					Yes					
Driveline Optimization			No					No					
GSA (Gear Shift Advisor)			Yes					Yes					
FE Details		MSA (Mode Shift Advisor)			No					No			
	Low CRR Tyres			Yes					Yes				
	Low Viscous rear axle oil (80W90)			Yes					Yes				
	Smart Alternator (LIN)			No					No				
	THU/Unitised/Low friction bearing			NO					No				
	Route			Jamshedpur-Ranchi-Jamshedpur					Jamshedpur-Ranchi-Jamshedpur				
	FE Duty Cycle (Laden/Unladen)			Laden Trial @ 40 kmph					Laden Trial @ 40 kmph				
				Laden Trial @ 40 Kmph					Laden Trial (With AC ON) @ 40 Kmph				
	Trial Date			04.05.2025	07.05.2025	05.05.2025	06.05.2025	08.05.2025	04.05.2025	07.05.2025	05.05.2025	06.05.2025	08.05.2025
	Trip Distance			242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	
	Total Diesel Quantity			64.08	62.41	62.97	64.70	64.41	65.36	63.88	66.71	67.85	
	Total Urea Quantity			4.40	4.00	4.00	4.00	4.50	4.00	3.20	4.20	4.30	
	Equivalent Urea Quantity : 50% Consumption			2.20	2.00	2.00	2.00	2.25	2.00	1.60	2.10	2.15	
	Total Eq. Fluid Consumed			66.28	64.41	64.97	66.70	66.66	67.36	65.48	68.81	70.00	
	Urea uses % wrt. Diesel			7%	6%	6%	6%	7%	6%	5%	6%	6%	
Fuel Economy			3.78	3.88	3.84	3.74	3.76	3.70	3.79	3.63	3.57		
Status of Fuel economy			%	Base	Base	Base	Base	-1.96%	-2.30%	-5.61%	-4.64%		
Equivalent Fluid Economy			kmpl	3.65	3.76	3.72	3.63	3.63	3.59	3.70	3.52	3.46	
Status of Equivalent Fluid economy			%	Base	Base	Base	Base	-1.60%	-1.63%	-5.58%	-4.71%		
Cluster Total Odo. Reading (at End of trial)			km	234	234	233	233	233	235	254	253	253	
Cluster Trip AFE			kmpl	3.70	3.80	3.80	3.80	3.70	3.90	4.00	3.80	3.80	
Avg.amp Results	Average Fuel Economy			kmpl	3.83		3.78		3.75		3.57		
	Status of Average Fuel economy			%	Base		Base		-2.13%		-5.49%		
	Average Eq. Fluid Economy			kmpl	3.76		3.66		3.70		3.47		
	Status of Average Eq. Fluid economy			%	Base		Base		-1.63%		-5.33%		
	Status of % Delta of AC ON wrt AC OFF						Base				-3.70%		

9.0 Torque table:-												
➤ Balance mode (TF – 130%) – Test Vehicle												
	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0
➤ Eco mode (TF – 120%) – Test Vehicle												
	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

10.1 Further Testing: As per further development plan.												
FIR Status											Page 4 of 4	

Annexure-1

Level 1 Attribute:		PANEL Chart		VE CVBU/FE/01								
PAT- Fuel Economy		Specification:Max Power- 220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD 5.12 RAR 295/90R20 Tyres iUSFE Torque Factor: E-130%, B-120% EOL Parameters : RSG: 80 kmph		Panel Chart No: VE CVBU/ HCV/ Signa 3523.T BSVI-II / FE/01								
Programme / MY : Signa 3523.T BSVI-II for AC regulation	Gateway : DR3	Owner: Ajaykumar Jadhav	Project Name: Signa 3523.T BSVI-II AC ON	Date: 20/05/2025								
Level 1 Summary:		Statement Of Intent (SOI) or DNA Description:										
Level 1:	<table><tr><td rowspan="2">PALS Position</td><td colspan="2">Status</td></tr><tr><td>Current/ Gateway</td><td>Job#1 Predicted</td></tr><tr><td>L</td><td>--</td><td></td></tr></table>		PALS Position	Status		Current/ Gateway	Job#1 Predicted	L	--		Fuel Economy: Haulage Application	
	PALS Position	Status										
		Current/ Gateway	Job#1 Predicted									
L	--											

Deriv.	Attribute Target Summary:											Remarks
	Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	Target	Status Current/Gateway SIGNA 3523.T BSVI -II AC ON	Job#1 Predicted	Base Vehicle Base SIGNA 3523.T BSVI -II AC OFF	
	Specification								LBSC - 1900,2000,35000 Kg VAM- 3.0 Kmph/s, 2.5Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph		LBSC - 1900,2000,35000 Kg VAM- 3.0 Kmph/s, 2.5Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph	1. Signa 3523.T BSVI-II Test vehicle with AC ON & Signa 3523.T BSVI-II base vehicle with AC OFF were run back to back with same configuration at 40kmph to get the AC ON delta % drop. 2. Signa 3523.T BSVI-II AC ON data extrapolated by applying AC ON % delta drop. 3. Target has been taken at-par wvt to base SIGNA 3523.T BSVI-II AC OFF vehicle. 4. AC PALS position to be carried forward from existing Non AC PALS position.
	Vehicle GVW (Payload)						kg		35000		35000	
	Power (Max) @ rpm						HP		220 hp @ 2300 rpm		220 hp @ 2300 rpm	
	Power to Weight ratio						HP/Ton		6.28		6.28	
	Torque (Max) @ rpm						Nm		925 Nm @ 1000-1300 rpm		925 Nm @ 1000-1300 rpm	
	Torque to Weight ratio						Nm/Ton		26.42		26.42	
	Tyres size						--		Bridgestone 295/90R20		Bridgestone 295/90R20	
	Gear Box Ratios						--		G950 DD, FGR -9.36		G950 DD, FGR -9.36	
	Final Drive Ratio						--		5.12		5.12	
	Fluid Economy at Rated Load						L		--			
	Highway AC ON						--					
	Route : Jamshedpur-Ranchi						--					
	Avg. Fluid Economy @40 kmph						kmpl	≥ 3.60	3.47	3.60		
	Avg. Fluid Economy @55 kmph						kmpl	-	3.27	3.40		
	Fluid Economy At Unladen					C		--				
	Highway AC ON						--					
	Route : Jamshedpur-Ranchi						--					
	Avg. Fluid Economy @40 kmph						kmpl	≥ 6.53	6.29	6.53		
	Avg. Fluid Economy @55 kmph						kmpl	-	5.68	5.90		
	Overall Fuel Economy (80% Laden + 20% Unladen)					L		--				
	Highway AC ON						--					
	Route : Jamshedpur-Ranchi						--					
	Overall Avg. Fluid Economy @40 kmph						kmpl	≥ 3.96	3.81	3.96		
	Overall Avg. Fluid Economy @55 kmph						kmpl	-	3.58	3.72		

Executive Summary:

1. Signa 3523.T BSVI-II Test vehicle with AC ON & Cowl 3523.T BSVI-II base vehicle with AC OFF were run back to back with same configuration at 40kmph to get the AC ON delta % drop.
2. Signa 3523.T BSVI-II AC ON data extrapolated by applying AC ON % delta drop.
3. Target has been taken at-par wrt to base SIGNA 3523.T BSVI -II AC OFF vehicle.
4. AC PALS position to be carried forward from existing Non AC PALS position.

Tested by:




Palash Das/ Swapnil Jadhav
PAT ENGINEER

Verified and Approved by:




Ajaykumar Jadhav / Sanjeev Kumar
PAT LEAD / PROJECT LEAD

 TATA MOTORS LTD. ERC,PUNE		PAT ATTRIBUTES SHEET - Fuel Economy										VE CVBU/FE/01							
		Project Name: Signa 3523.T BSVI-II (AC ON) Max Power- 220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD 5.12 RAR 295/90R20 Tyres iUSFE Torque Factor: E-130%, B-120% Base Vehicle: Signa 3523.T BSVI-II AC OFF Competitor Vehicle: Application: Haulage										PAT Sheet No.: VE CVBU/ HCV/ VE CVBU/ HCV/ Signa 3523.T BSVI-II / FE/01							
Depart ment Name: Vehicle Engineering												Date: 20/05/2025							
Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification @DR4	Ref.Standard / Work Guideline	Simulation	Base Vehicle	Target	Test Vehicle	Status (RYG)	Remarks	CoC Remark		
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7												SIGNA 3523.T BSVI-II AC OFF (G 950 DD RAR 5.12 295/90R20)	SIGNA 3523.T BSVI-II AC ON (G 950 DD RAR 5.12 295/90R20)
A	Fuel Economy Evaluation						Heading						LBSC - 1900,2000,35000 Kg VAM- 3.0 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph	LBSC - 1900,2000,35000 Kg VAM- 3.0 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph	--				
2.6	Fuel Economy At Rated Load						Heading												
2.7	Highway AC ON						Heading												
2.7.1	Route : Jamshedpur-Ranchi						Kmpl	Standard	Specification	TST/VTWG/IS8									
2.7.9	Average Fuel Economy @40 kmph								Data Generation			3.60	--		--				
2.7.10	Average Fuel Economy @55 kmph								Data Generation			3.38	--		--				
2.7.11	Average Fluid Economy @40 kmph								Target			3.60	≥ 3.60	3.47	R				
2.7.12	Average Fluid Economy @55 kmph								Data Generation			3.40	--	3.27	--				
4.6	Fuel Economy At Unladen						Heading												
4.7	Highway AC ON						Heading												
4.7.1	Route : Jamshedpur-Ranchi						Kmpl	Standard	Specification	TST/VTWG/IS8									
4.7.9	Average Fuel Economy @40 kmph								Data Generation			6.16	--		--			1. Signa 3523.T BSVI-II Test vehicle with AC ON & Signa 3523.T BSVI-II base vehicle with AC OFF were run back to back with same configuration at 40kmph to get the AC ON delta % drop.	
4.7.10	Average Fuel Economy @55 kmph								Data Generation			5.74	--		--			2. Signa 3523.T BSVI-II AC ON data extrapolated by applying AC ON % delta drop.	
4.7.11	Average Fluid Economy @40 kmph								Target			6.53	≥ 6.53	6.29	R			3. Target has been taken at par wrt to base SIGNA 3523.T BSVI-II AC OFF vehicle.	
4.7.12	Average Fluid Economy @55 kmph								Data Generation			5.90	--	5.68	--				
5.0	Overall Fuel Economy (80% Laden + 20% Unladen)						Heading												
5.1	Highway AC ON						Heading												
5.1.1	Route : Jamshedpur-Ranchi						Kmpl	Standard	Specification	TST/VTWG/IS8									
5.1.9	Average Overall Fuel Economy @40 kmph								Data Generation			3.93	--		--				
5.1.10	Average Overall Fuel Economy @55 kmph								Data Generation			3.68	--		--				
5.1.11	Overall Average Fluid Economy @40 kmph								Target			3.96	≥ 3.96	3.81	R				
5.1.12	Overall Average Fluid Economy @55 kmph								Data Generation			3.72	--	3.58	--				
Sign Off												Index for Status							
Responsibility		Name				Signature				Off target, no resolution									
PAT Engineer		Palash Das/Swapnil Jadhav				 				Marginally off target, resolution planned/low risk									
PAT Lead / Project Lead		Ajaykumar Jadhav / Sanjeev kumar				 				Met target, no concern									

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

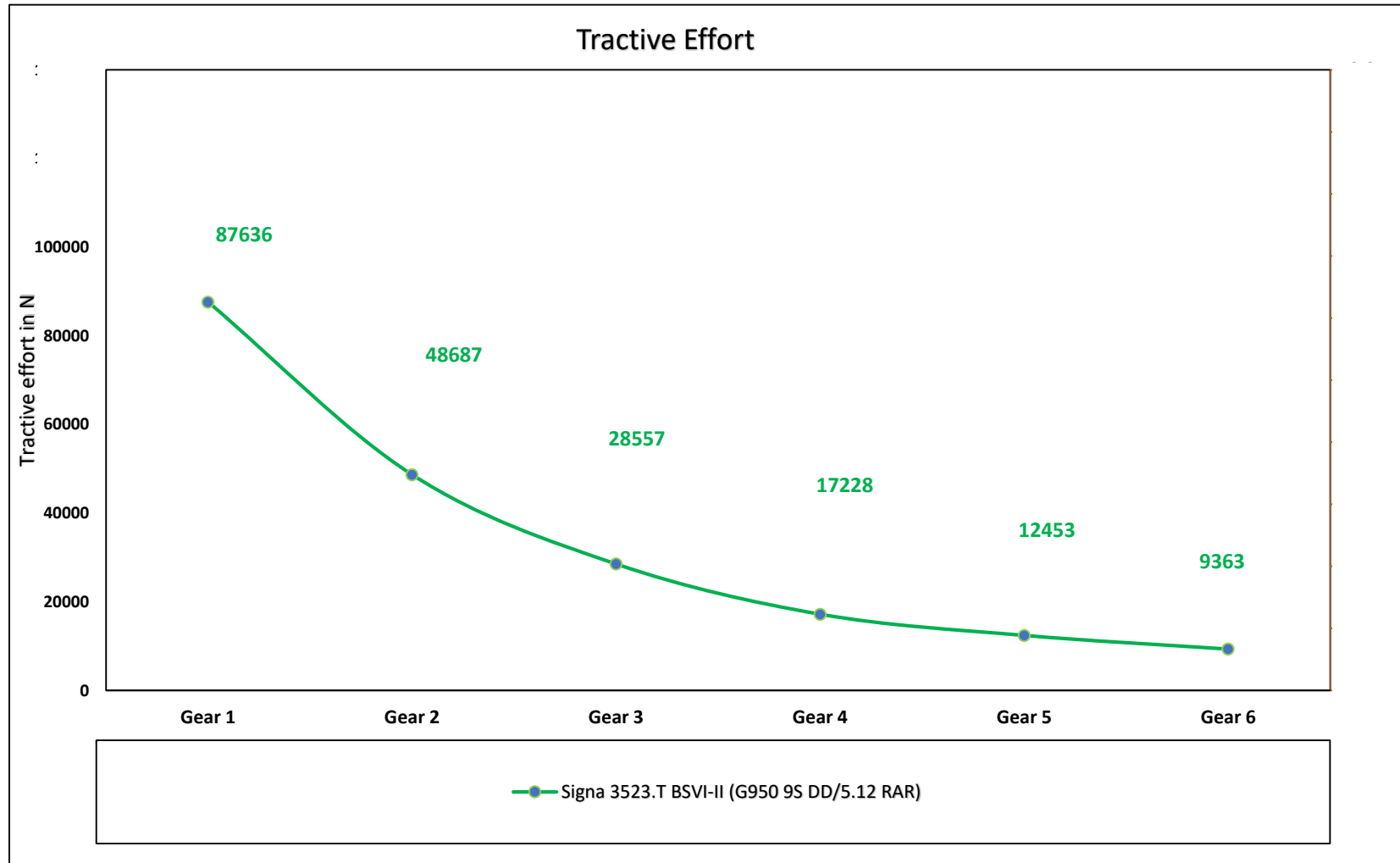
Executive FE summary of Signa 3523.T BSVI-II vehicle with AC ON in comparison with Cowl 3523.T BSVI-II Base Vehicle with AC OFF.

PED | HCV | 20.05.2025



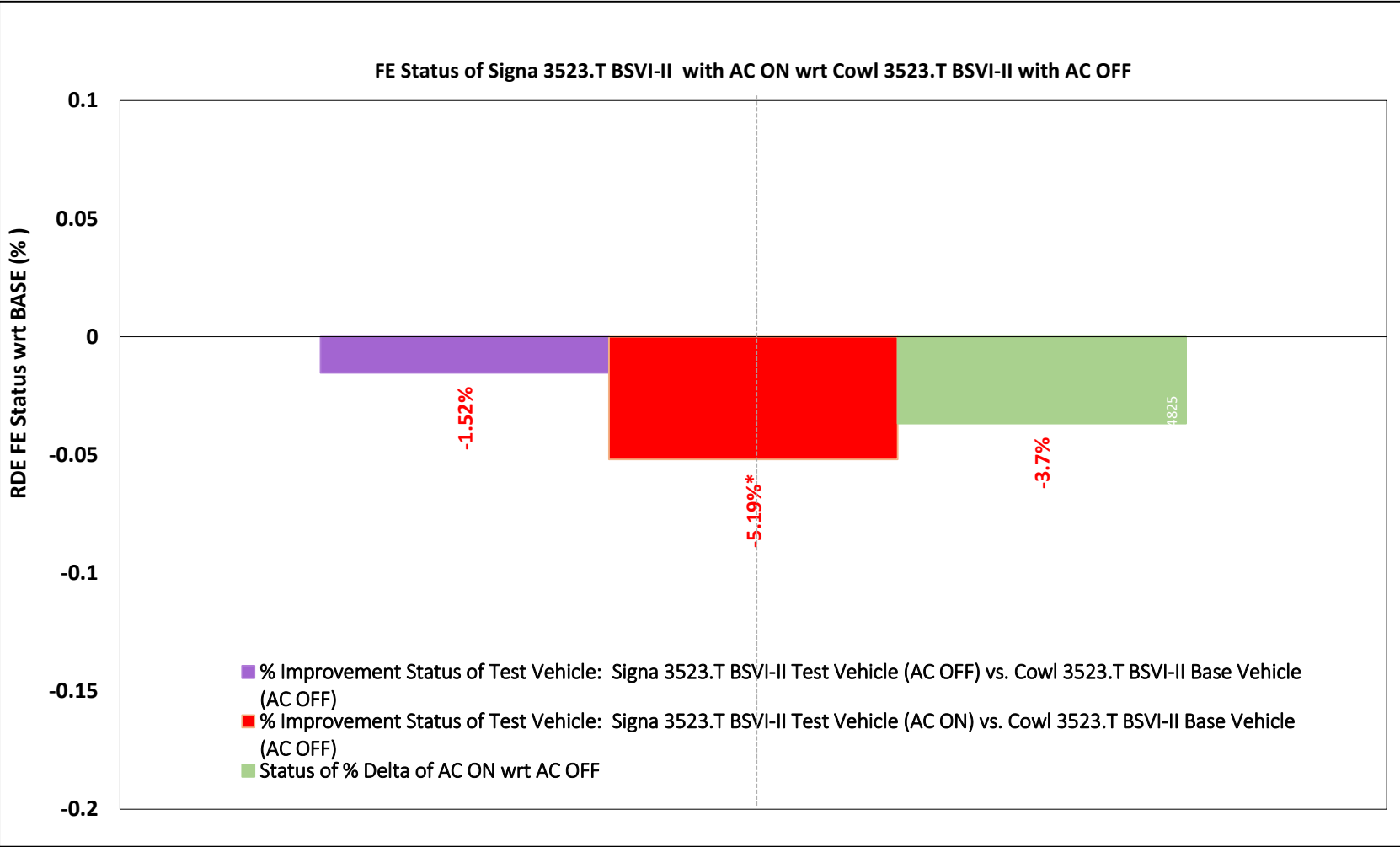
Sr. No.	Parameters	Base Vehicles BSVI-II	Test Vehicles BSVI-II
		Cowl 3523.T RDE BSVI	Signa 3523.T RDE BSVI
1	Engine	6 Cylinder 5.6 L / 230 HP BSVI	6 Cylinder 5.6 L / 230 HP BSVI
1.1	Engine Max. Power	220 HP	220 HP
1.2	Engine Max. Torque	925Nm @ 1000-1300rpm	925Nm @ 1000-1300rpm
1.3	Power to weight (HP/Ton)	6.28	6.28
1.4	Torque to weight (Nm/Ton)	26.42	26.42
1.5	Engine Calibration Number	XWTAA19D07	XWTAA19D07
1.6	Engine Calibration Number (Production)	PF10040	PF10040
1.7	Engine Calibration Maturity	100%	100%
1.8	Regeneration Strategy	Auto	Auto
1.9	Urea Dosing	Yes	Yes
2.1	EOL Parameters	LBSC - 1900,2000,35000 kg, VAM - 3.0,2.5, 10, 40 Kmph, GDP - 50, 50 Kmph, RSG – 80 Kmph	LBSC - 1900,2000,35000 kg, VAM - 3.0,2.5, 10, 40 Kmph, GDP - 50, 50 Kmph, RSG – 80 Kmph
2	Fuel Economy Improvement Technologies		
2.1	Engine Cooling Fan	E-Viscous	E-Viscous
2.2	Fuel Economy Switch	4TC	4TC
2.3	Gear Shift Adviser	Yes	Yes
2.4	Accelerator Pedal Map	Yes	Yes
3	Other Parameters		
3.1	Fuel Tank type	Polymer	Polymer
3.2	Driveline	G 950 DD , FGR 9.36	G 950 DD , FGR 9.36
3.3	Gear Ratio	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,
3.4	RAR	5.12	5.12
3.5	Tire size	Front – Rib 295/90R 20 Bridgestone Tyre Rear – Lug 295/90R20 Bridgestone Tyre	Front – Rib 295/90R 20 Bridgestone Tyre Rear – Lug 295/90R20 Bridgestone Tyre

Tractive Effort Comparison :



Tractive effort of Signa 3523.T BSVI-II vehicle

Fuel Economy Status



FE Status of Signa 3523.T BSVI-II wrt Cowl 3523.T BSVI-II Base Vehicle.

Fuel Economy summary : AC ON & OFF

Parameters			Base Vehicle : OBD-2					Test Vehicle : OBD-2				
Sr. No.			LPT 3523.T Tata RDE BSVI					SIGMA 3523.T Tata RDE BSVI				
1	Vehicle Identification No. (ERC No./EJ No./Reg. No.)		ERC-1310					ERC-1309				
2	Vehicle GVW/GCW	kg	35000					35000				
3	Engine Details (Make/cc/Norms/no. of Cyl.)		Cummins ISBe 5.6L/ Ashwamedh					Cummins ISBe 5.6L/ Ashwamedh				
	Engine Max Power		220 hp @ 2300 rpm					220 hp @ 2300 rpm				
	Engine Max Torque		925 Nm @ 1000-1300 rpm					925 Nm @ 1000-1300 rpm				
4	Engine Calibration Maturity Level		100% Cal Maturity					100% Cal Maturity				
	Regeneration Strategy (Time ase/Soot base)		Time Based (100 Hrs)					Time Based (100 Hrs)				
5	Gear Box (make/drive/FGR)		TATA G-950 DD					TATA G-950 DD				
6	Gear Ratios		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00					1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00				
	RAR		5.12					5.12				
7	Tyre Details		295/90R 20 RIB Bridgestone Low LCRR					295/90R 20 RIB Bridgestone Low LCRR				
8	LBSC	High & Low Engine Speed Breakpoint	RPM	1900 rpm, 2000 rpm				1900 rpm, 2000 rpm				
	VAM	Vehicle Mass Threshold	kg	35000				35000				
		Variable Acceleration 1, 2	kmph/sec	3.0,2.5				3.0,2.5				
	GDP	Variable Acceleration Speed 1, 2	kmph	10, 40				10, 40				
		GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	50, 50				50,50				
	RSG	Maximum Accelerator Vehicle Speed	kmph	80				80				
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	120%				120%				
		Torque Limit Maximum Adjustment Factor	Economy	130%				130%				
	PTP	Powertrain Protection (Enable / Disable)		Disable				Disable				
		Torque Limit		NA				NA				
9	USFE / Torque Table Revision (1/2/3/n)		USFE-1					USFE-1				
FE Avenues (Engine Level)	Fan Details (type/make/blades)		24" High Efficiency Viscous fan					24" High Efficiency Viscous fan				
	Base Cal Improvement (Cal Maturity %)		100% Cal Maturity					100% Cal Maturity				
	Low Viscous Engine Oil (TMGO-10W30)		Yes (15W40CK4+)					Yes (15W40CK4+)				
	FE Switch		N/A					N/A				
	Engine Features (EOL : Enable/Disable)		Enable					Enable				
	ATS Optimization		Yes					Yes				
	APM		Yes					Yes				
	Cruise Control		No					No				
	Gear based VAM		No					No				
	Intelligent mode shifter		No					No				
FE Avenues (Vehicle Level)	Moving Mass Estimation		Yes					Yes				
	Driveline Optimization		No					No				
	GSA (Gear Shift Advisor)		Yes					Yes				
	MSA (Mode Shift Advisor)		No					No				
	Low CRR Tyres		Yes					Yes				
	Low Viscous rear axle oil (80W90)		Yes					Yes				
	Smart Alternator (LIN)		No					No				
	THU/Unitised/Low Friction bearing		No					No				
	Route		Jamshedpur-Ranchi-Jamshedpur					Jamshedpur-Ranchi-Jamshedpur				
	FE Trial Details	FE Duty Cycle (Laden/Unladen)		Laden Trial @ 40 kmph		Laden Trial @ 40 Kmph			Laden Trial @ 40 kmph		Laden Trial (With AC ON) @ 40 Kmph	
Trial Date		04.05.2025	07.05.2025	05.05.2025	06.05.2025	08.05.2025	04.05.2025	07.05.2025	05.05.2025	06.05.2025	08.05.2025	
Trip Distance		km	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	
Total Diesel Quantity		Litre	64.08	62.41	62.97	64.70	64.41	65.36	63.88	66.71	67.85	
Total Urea Quantity		Litre	4.40	4.00	4.00	4.00	4.50	4.00	3.20	4.20	4.30	
Equivalent Urea Quantity : 50% Consumption		Litre	2.20	2.00	2.00	2.00	2.25	2.00	1.60	2.10	2.15	
Total Eq. Fluid Consumed		Litre	66.28	64.41	64.97	66.70	66.66	67.36	65.48	68.81	70.00	
Urea uses % wrt. Diesel		%	7%	6%	6%	6%	7%	6%	5%	6%	6%	
Fuel Economy		kmpl	3.78	3.88	3.84	3.74	3.76	3.70	3.79	3.63	3.57	
Status of Fuel economy		%	Base	Base	Base	Base	Base	-1.96%	-2.30%	-5.61%	-4.64%	
Equivalent Fluid Economy		kmpl	3.65	3.76	3.72	3.63	3.63	3.59	3.70	3.52	3.46	
Status of Equivalent Fluid economy		%	Base	Base	Base	Base	Base	-1.60%	-1.63%	-5.58%	-4.71%	
Cluster Total Odo. Reading (at End of trial)		km	234	234	233	233	233	235	254	253	253	
Cluster Trip AFE		kmpl	3.70	3.80	3.80	3.80	3.70	3.90	4.00	3.80	3.80	
Avg.kmpl Results		Average Fuel Economy		kmpl	3.83		3.78			3.75		3.57
	Status of Average Fuel economy		%	Base		Base			-2.13%		-5.49%	
	Average Eq. Fluid Economy		kmpl	3.76		3.66			3.70		3.47	
	Status of Average Eq. Fluid economy		%	Base		Base			-1.63%		-5.33%	
Status of % Delta of AC ON wrt AC OFF			Base					-3.70%				

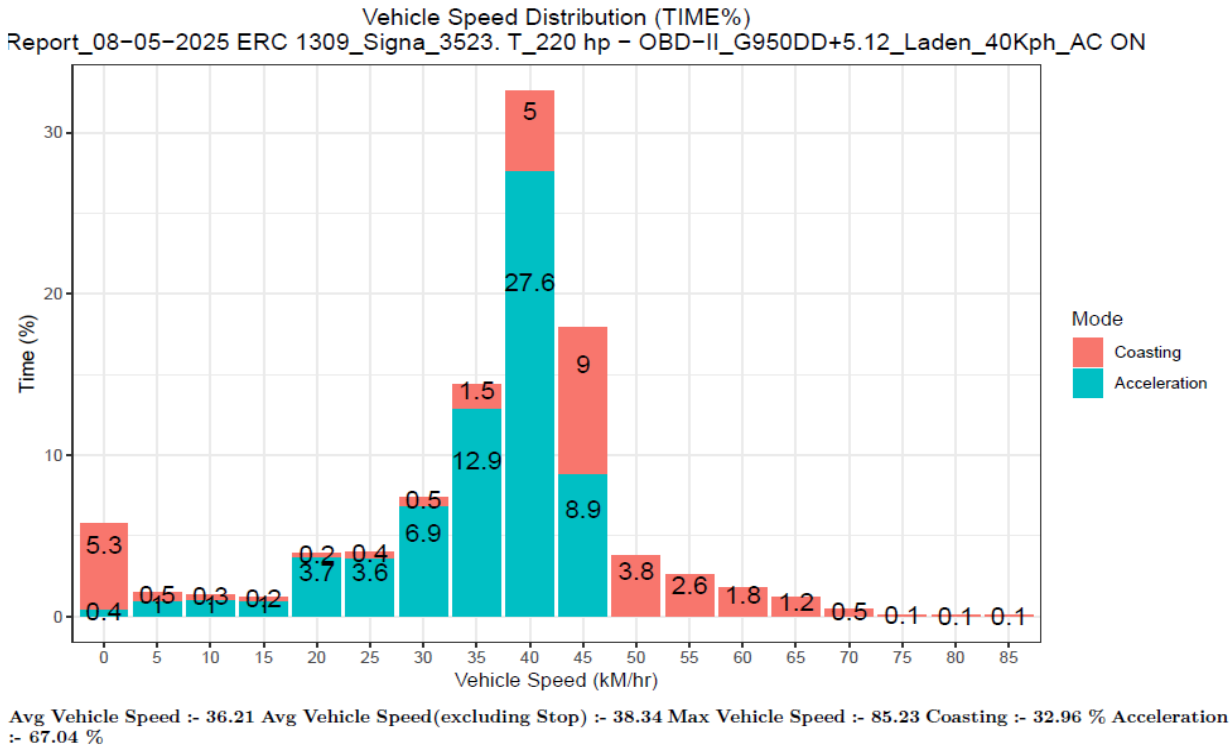
Duty Cycle Analysis –Jamshedpur-Ranchi highway FE Trial

Duty Cycle Summary (AC ON): Laden (40 kmph)

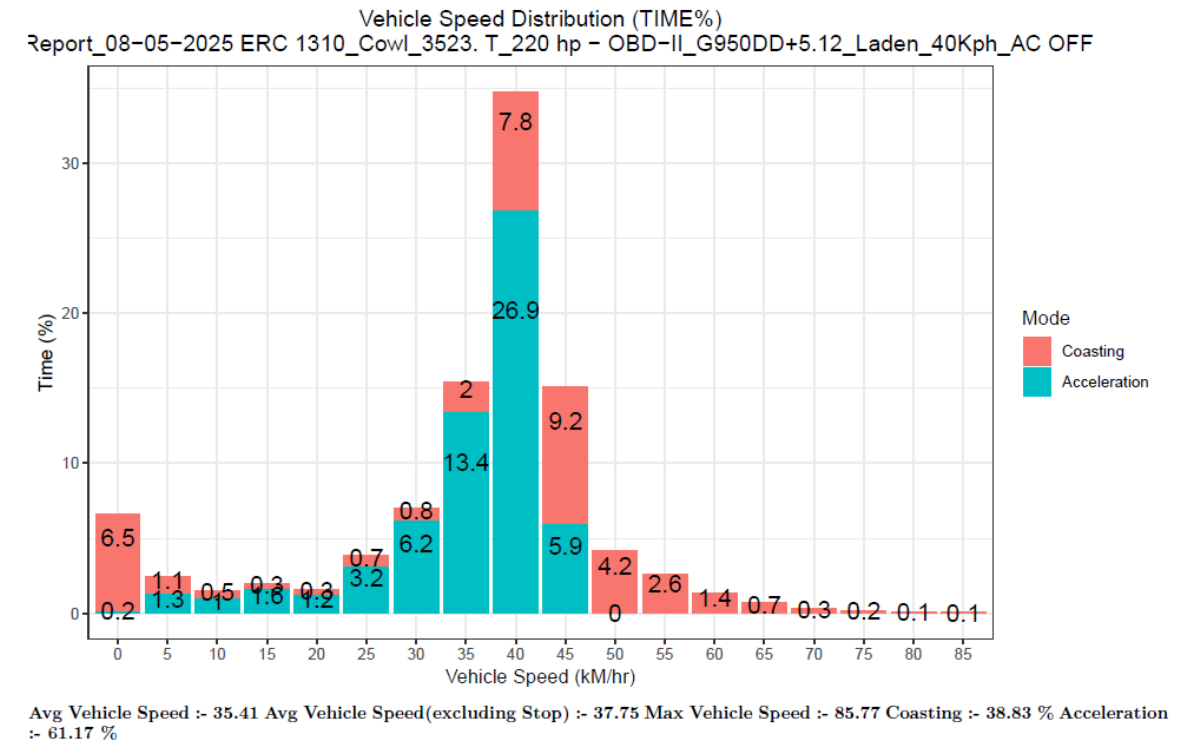
		(40 kmph)	
Parameters		Signa 3523.T RDE BSVI-II Test Vehicle	Signa 3523.T RDE BSVI-II Base Vehicle
Engine idle speed (rpm)		650	650
Low idle time (%)		5.27	6.28
Average engine speed (rpm)		1095	1060
Average power (kW)		36	35
Avg. vehicle speed (kmph)	Including Stop	38	38
	Excluding Stop	40.77	38.81
FE switch utilization (%time)	Power	23.64	21.10
	Balance	0	0
	Eco	76.36	78.90
Gear utilization (%time)	Neutral Gear	5.7	6.3
	5 th Gear	5.00	5.00
	6 th Gear	64.60	65.32
	Total shifts	302	314
Throttle (%time)	100%	21.8	28.6
	0%	32.2	37.8
Coolant Temp (°C)	80°C	0.1	10.4
	85°C	1	44.2
	90°C	43.5	27.9
Duty Cycle fuel economy (kmpl)		3.86	4.11
Duty cycle Distance(km)		199	243
Duty Cycle Time(Hrs)		5.46	6.86
Actual Avg. Fuel Economy		3.66	3.47

Vehicle Speed Distribution

Signa 3523.T RDE BSVI-II Test Vehicle at Laden 40 Kmph

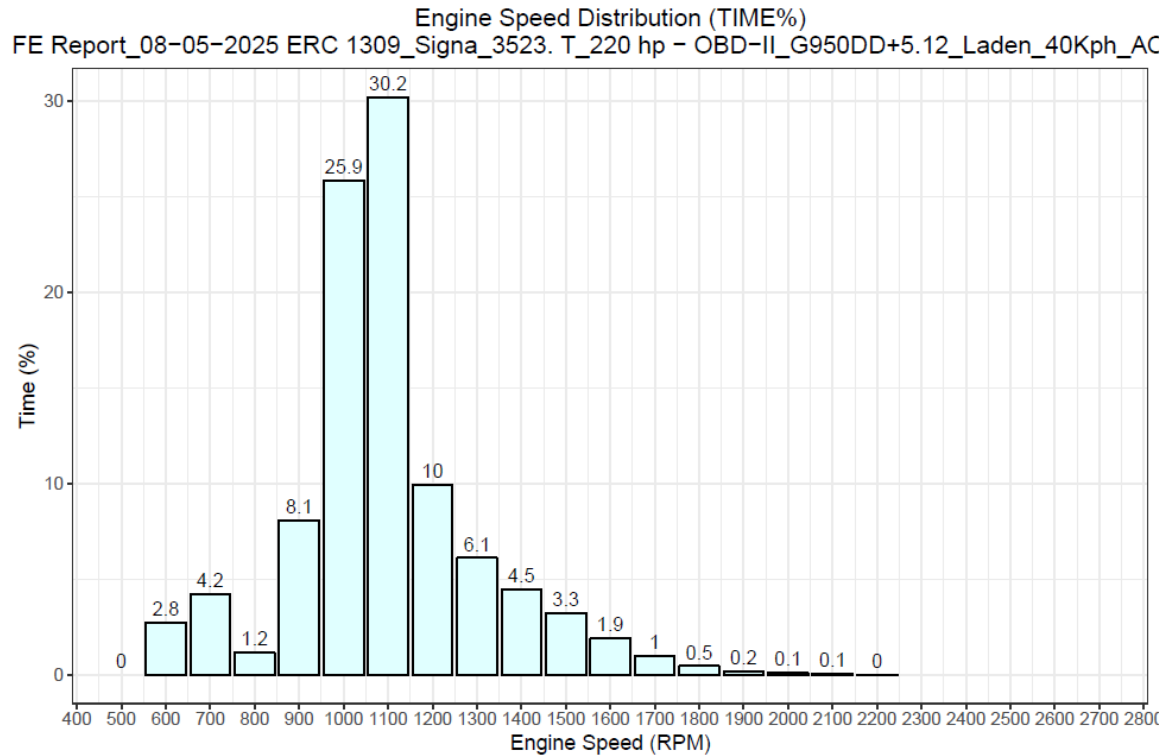


Cowl 3523.T RDE BSVI-II Base Vehicle at Laden 40 Kmph

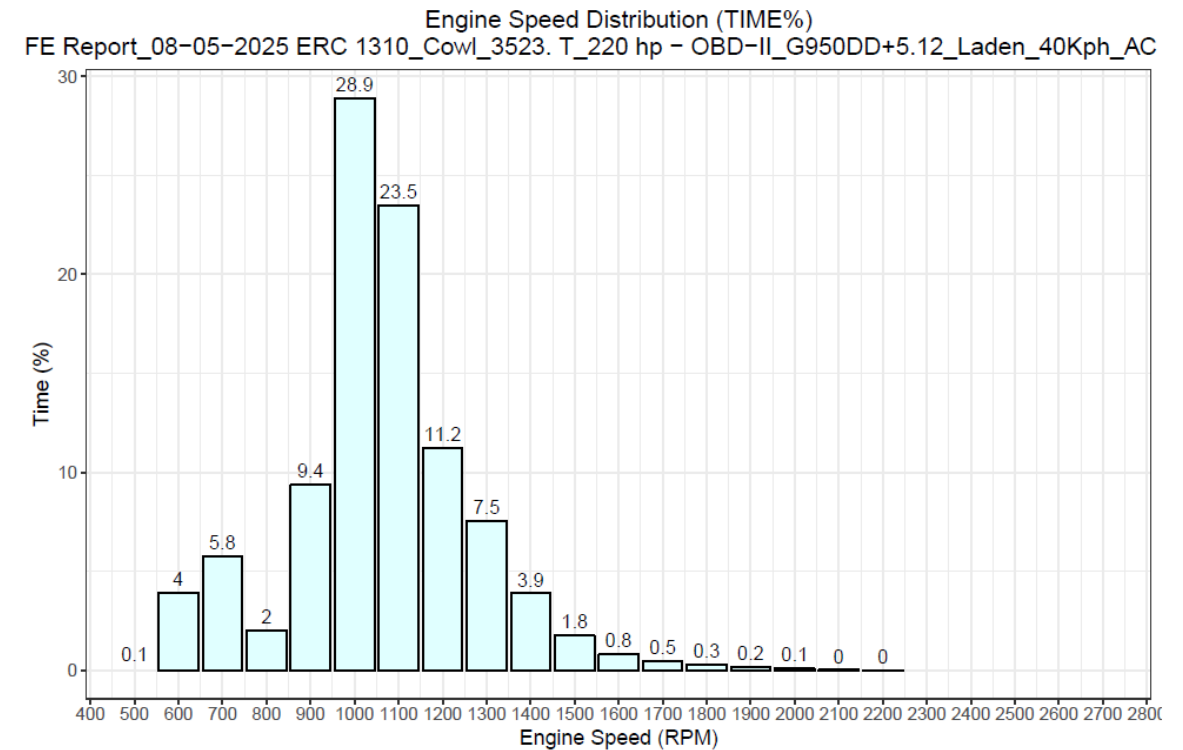


Engine Speed Distribution

Signa 3523.T RDE BSVI-II Test Vehicle at Laden 40 Kmph

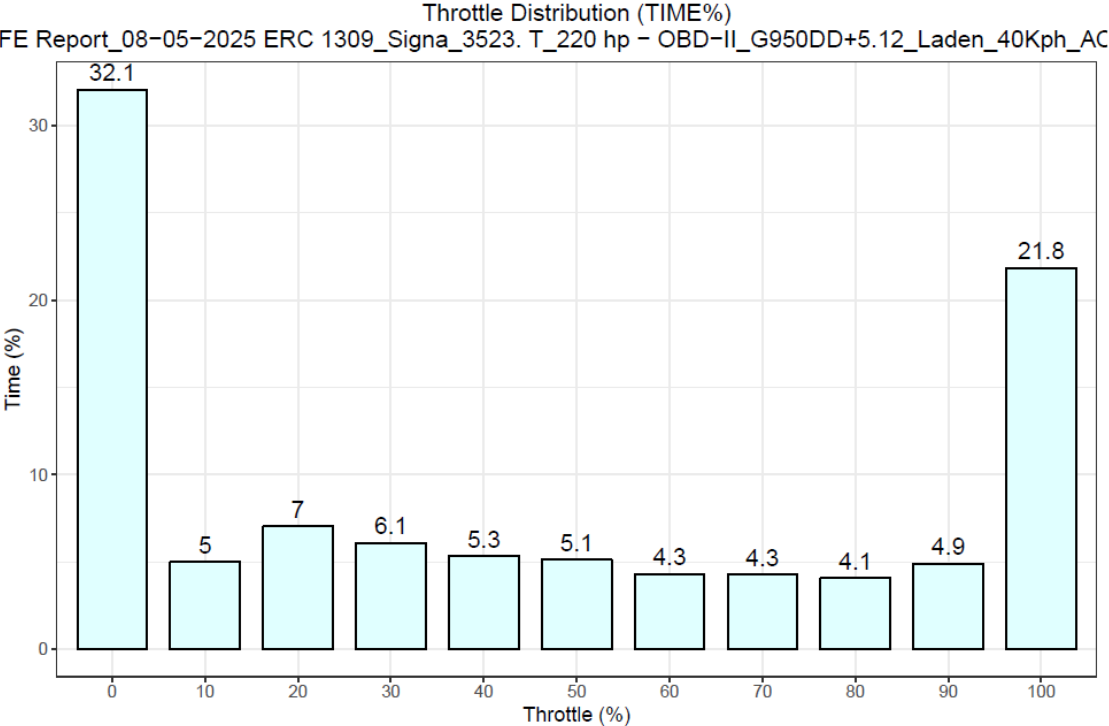


Cowl 3523.T RDE BSVI-II Base Vehicle at Laden 40 Kmph

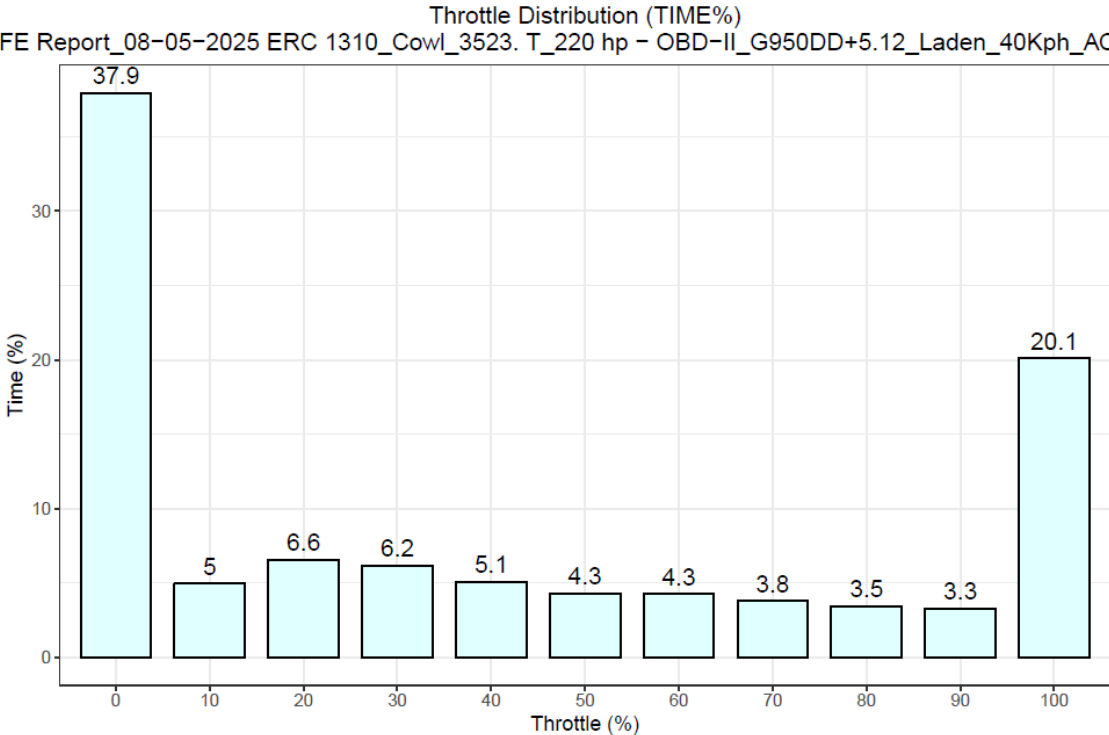


Throttle Distribution

Signal 3523.T RDE BSVI-II Test Vehicle at Laden 40 Km/h

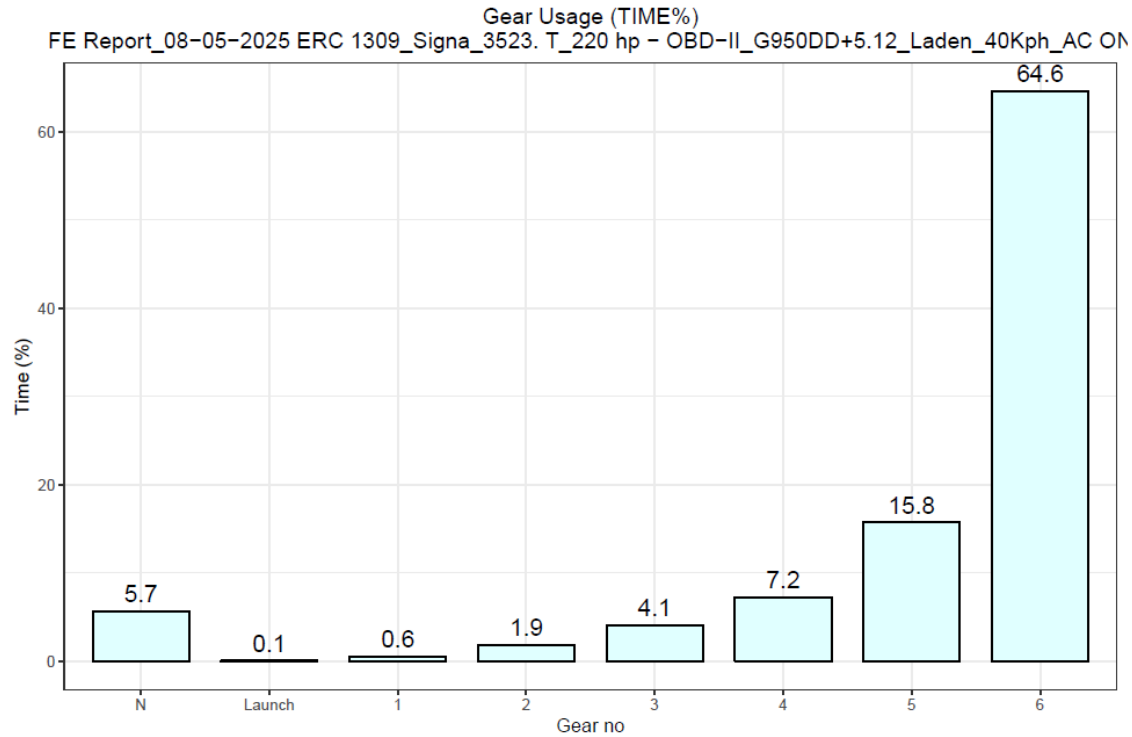


Cowl 3523.T RDE BSVI-II Base Vehicle at Laden 40 Km/h

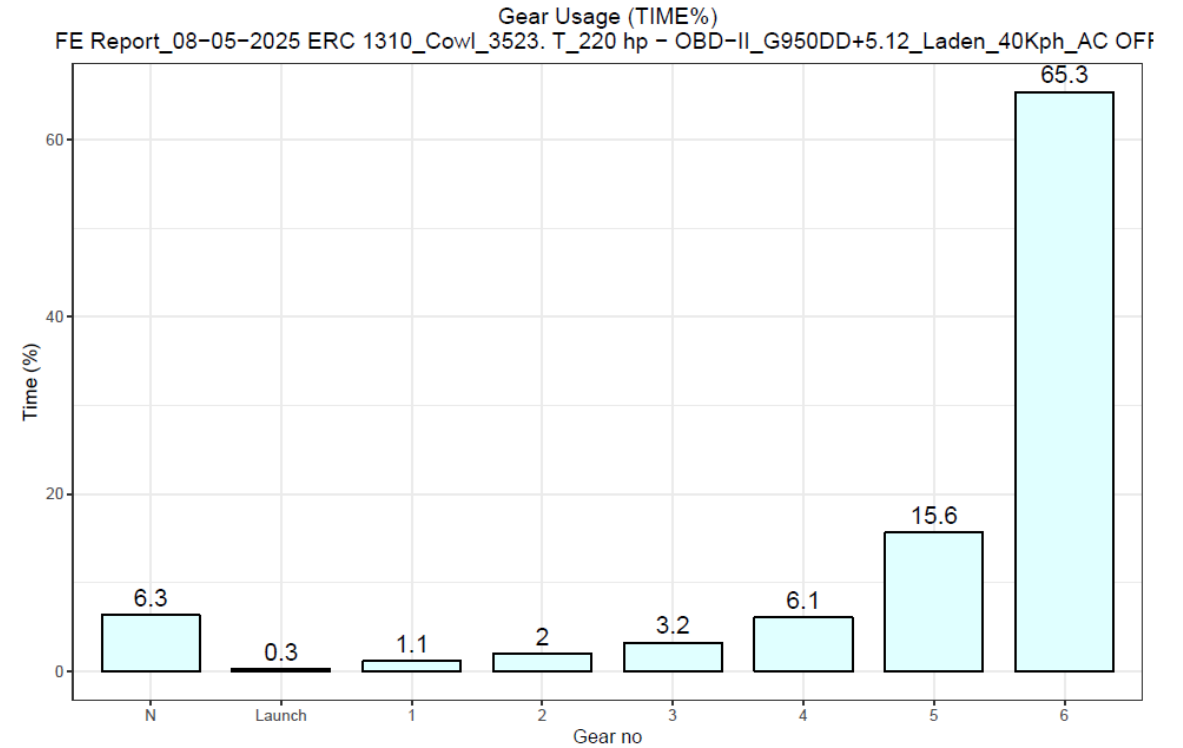


Gear Utilization

Signal 3523.T RDE BSVI-II Test Vehicle at Laden 40 Km/h



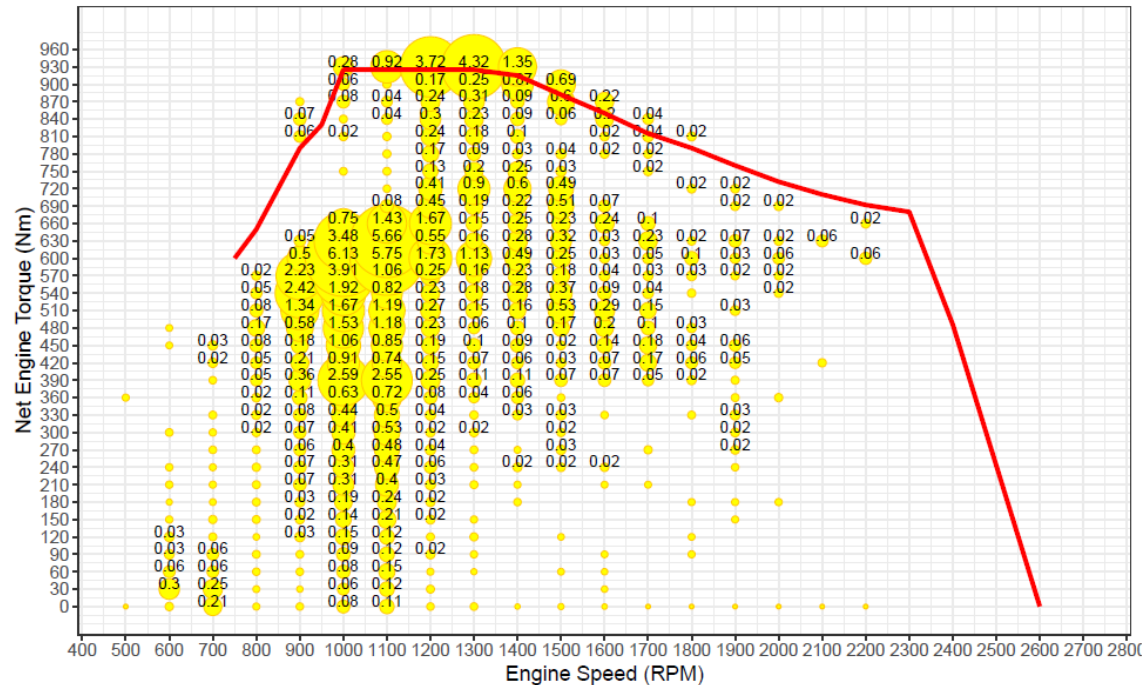
Cowl 3523.T RDE BSVI-II Base Vehicle at Laden 40 Km/h



Fuel spent % as Engine speed & torque

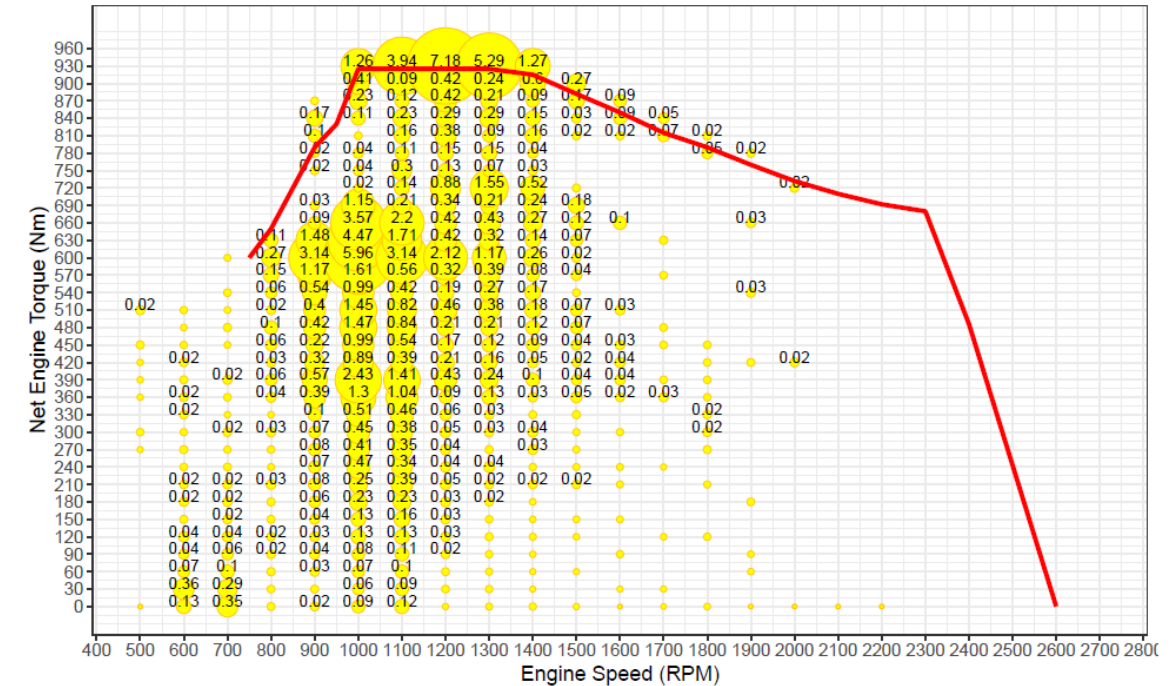
Signa 3523.T RDE BSVI-II Test Vehicle at Laden 40 Kmph

Fuel Index - Economy Mode :- 76.36 Power Mode :- 23.64 Balance Mode :- 0
FE Report_08-05-2025 ERC 1309_Signa_3523. T_220 hp - OBD-II_G950DD+5.12_Laden_40Kph_AC



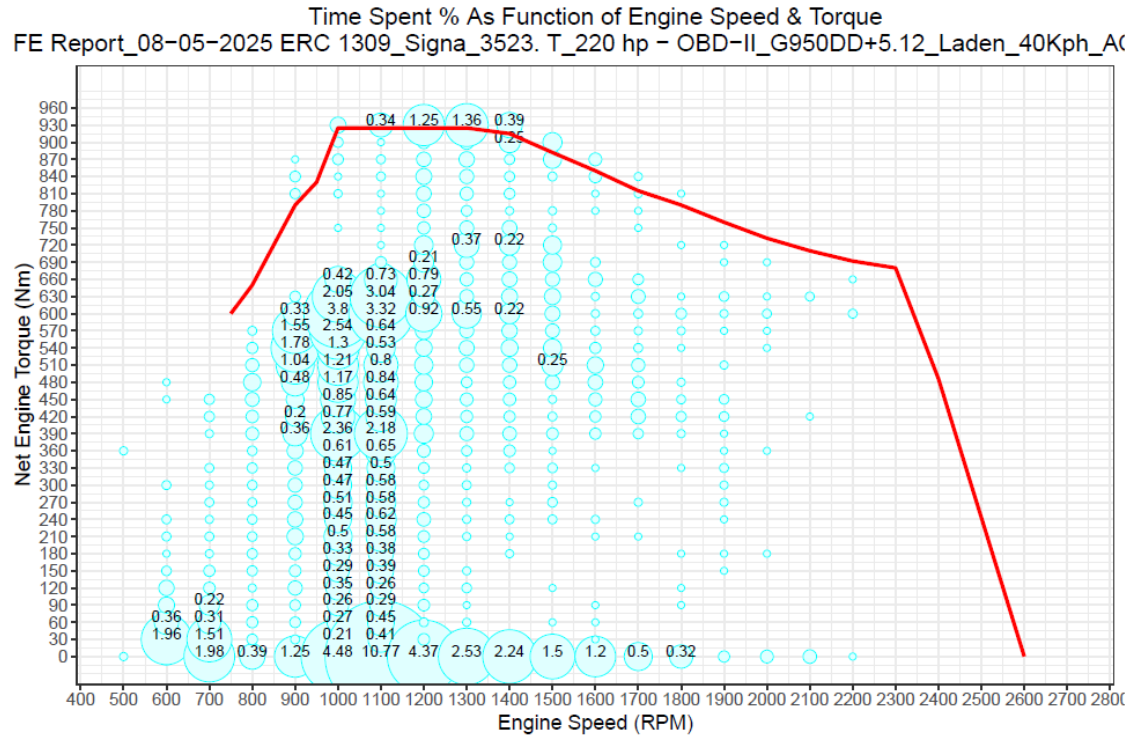
Cowl 3523.T RDE BSVI-II Base Vehicle at Laden 40 Kmph

Fuel Index - Economy Mode :- 78.9 Power Mode :- 21.1 Balance Mode :- 0
FE Report_08-05-2025 ERC 1310_Cowl_3523. T_220 hp - OBD-II_G950DD+5.12_Laden_40Kph_AC

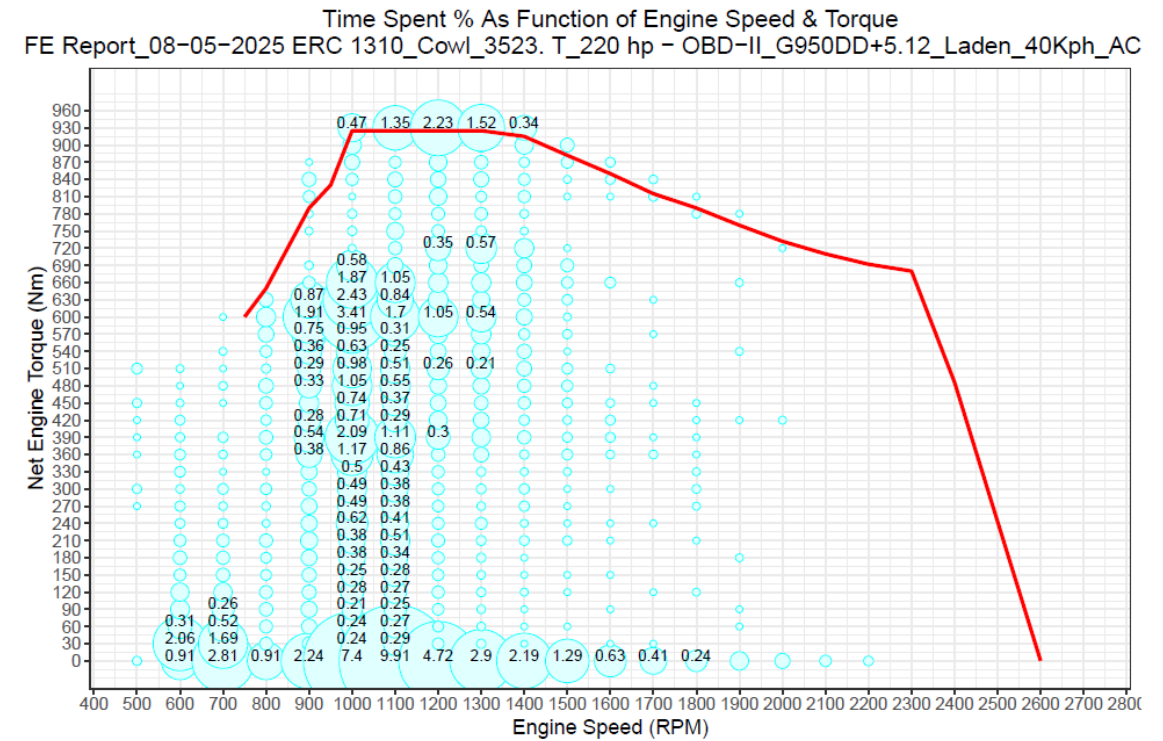


Time spent % as Engine speed & torque

Signa 3523.T RDE BSVI-II Test Vehicle at Laden 40 Kmph

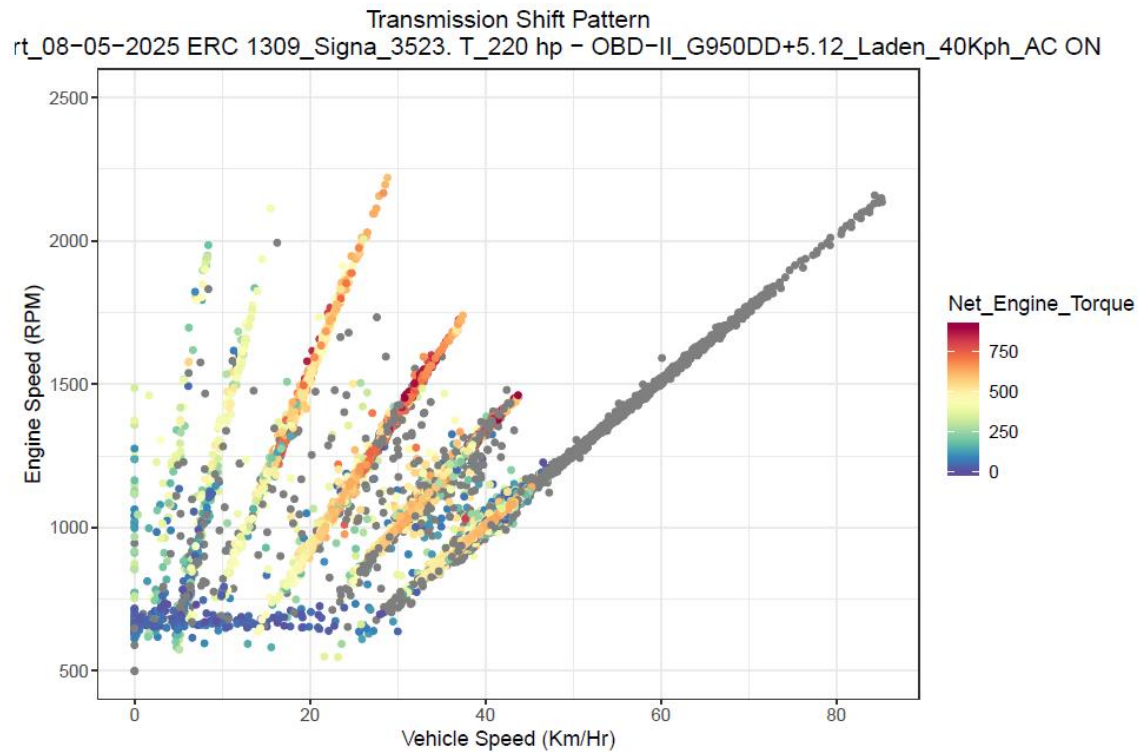


Cowl 3523.T RDE BSVI-II Base Vehicle at Laden 40 Kmph

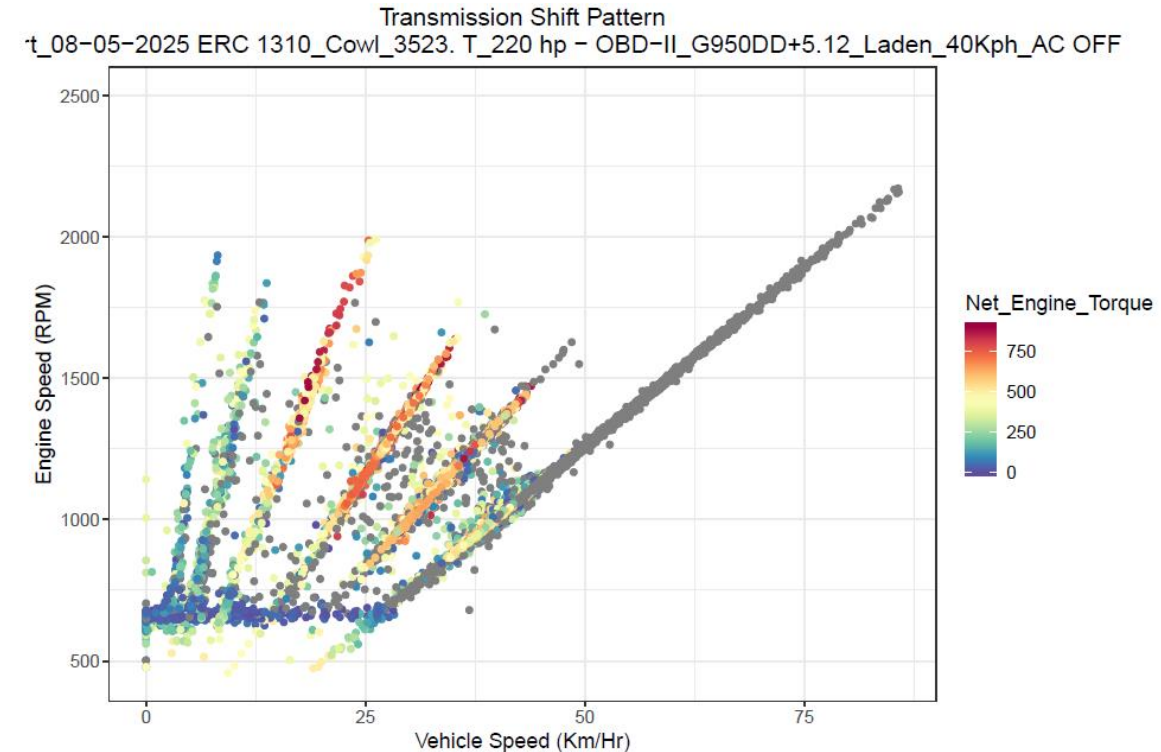


Transmission Shift Pattern

Signa 3523.T RDE BSVI-II Test Vehicle at Laden 40 Kmph



Cowl 3523.T RDE BSVI-II Base Vehicle at Laden 40 Kmph



Fuel Economy Comparison summary

Vehicle Model for X-axis	Crusing speed	Duty Cycle	FLE Status of Cowl 3523.T BSVI-II Base Vehicle with AC OFF	FLE Status of Signa 3523.T BSVI-II Test Vehicle with AC OFF	FLE Status of Cowl 3523.T BSVI-II Base Vehicle with AC OFF	FLE Status of Signa 3523.T BSVI-II Test Vehicle with AC ON	% Improvement Status of Test Vehicle: Signa 3523.T BSVI-II Test Vehicle (AC OFF) vs. Cowl 3523.T BSVI-II Base Vehicle (AC OFF)	% Improvement Status of Test Vehicle: Signa 3523.T BSVI-II Test Vehicle (AC ON) vs. Cowl 3523.T BSVI-II Base Vehicle (AC OFF)	Status of % Delta of AC ON wrt AC OFF
FE Status of Signa 3523.T BSVI-II with AC ON wrt Cowl 3523.T BSVI-II with AC OFF	40 kmph (Laden)	Laden	3.76	3.70	3.66	3.47	-1.52%	-5.19%	-3.7%

Thank You

TATA MOTORS LIMITED ERC PUNE	TEST PLAN		DATE: 20/05/2025			
	VEHICLE TESTING DEPARTMENT					
Description: Fuel Economy evaluation of Signa 3523.T BSVI-II for AC regulation			Test Item No: UA001188			
			New	✓	Moderate	✗
			Major	✗	Minor	✗

Part No: 52075256R,		Modification No: E;2,	Enclosures: Annexure 1-7
Drawing No: 52075256R, 520700000001			
Make: NA		WBS Number: P.24.EA.550- AC REGULATION-TRUCK	Copies to :
Vehicle Type: Request from Mr.Bimal Kishor dated 05/01/2023		No. of Samples: -	
Test Requested by:	Testing-Outdoor-UA/01326		Equipments:
Test Requisition Reference:			
Reason for Test/s:	Regulatory Requirement		
Regulatory Requirement:	a) Component: TST/VT/WG/158		
	b) Vehicle: NA		

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	Fuel Economy evaluation of Signa 3523.T BSVI-II for AC regulation	AS Per PED standard	TST/VT/WG/158	1	Palash Das	Beta	completed

REMARKS:		INTERNAL REF:	
TEST TO FAILURE: No			
MEOST: No			
RELIABILITY TESTING: No			
BENCHMARKING TESTING: No			
DFMEA/FMEA DONE: Yes			
PREPARED BY		CHECKED BY	APPROVED BY
Palash Das			

TEST REQUISITION

Ref.	SIGNA3523.T/FE/2025/02	Date:	27/04/2025
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3523.T BSVI			
2	Part Number	52075256R,			
3	Drawing number	52075256R, 520700000001	Mod status	E;2,	Mod Date 15/03/2025
4	Vehicle Model / Project	Request from Mr.Bimal Kishor dated 05/01/2023			
5	WBS Number	P.24.EA.550AC REGULATION-TRUCK			
6	Supplier	NA			
7	Number of samples	-			
8	Inspection report	Enclosed	✓	If No commitment date:	
9	Material report	Enclosed	✓	If No commitment date:	
10	Suppliers test report	Enclosed	✓	If No commitment date:	
11	Reason for testing	First Source Development	✗	Weight Reduction	✗
		Alternate source development	✗	Cost Reduction	✗
		Data generation Test	✗	Regulatory test	✓
		Any other (pl. specify)	✗	New Development	✗
12	What testing required?				
13	Remarks	FE testing of SIGNA 3523.T BSVI with AC			
14	Peripheral Parts Availability	Yes			

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

Authorized Signature:	
Name:	
Dept:	

Contact Person:	Kundan Kumar
Phone Number	

PALASH DAS [TMCV, Engineering, Jamshedpur]

From: MANORANJAN SAHU [TMCV, Engineering, Jamshedpur]
Sent: 21 May 2025 17:32
To: SANJEEV KUMAR [TMCV, Engineering, Jamshedpur]; AJAYKUMAR JADHAV [TMCV, Engineering, Pune]; PANKAJ HAWARE [TMCV, Engineering, Pune]
Cc: PAVAN SINDGIKAR [TMCV, Engineering, Pune]; VISHNU DIXIT [TMCV, Engineering, Jamshedpur]; RAVINDRA DESHMUKH [TMCV, Engineering, Pune]; PRANAB SOU MANDAL [TMCV, Engineering, Jamshedpur]; DEBABRATA DAS [TMCV, Engineering, Jamshedpur]; SWAPNIL JADHAV [TMCV, Engineering, Pune]; PALASH DAS [TMCV, Engineering, Jamshedpur]; RAVI RANJAN [TMCV, Engineering, Jamshedpur]; SUJEET PRASAD [TTL, ER&D TML_iSource_CVBU]; SUBHANKAR BANERJEE [TTL, ER&D TML_iSource_CVBU]; RANA SINGH [TMCV, Engineering, Jamshedpur]; JITENDRA KUMAR SINGH [TMCV, Engineering, Jamshedpur]; AMIT KHOT [TTL, ER&D TML_iSource_CVBU]; SANTOSH KUMAR [TMCV, Engineering, Pune]; NAROTTAM PANKAJ [TMCV, Engineering, Pune]; KUNDAN KUMAR [TMCV, Engineering, Jamshedpur]
Subject: RE: HCV Truck_FE/PD Beta Validation FIR/Test Report- All lead & follower models name to be mention
Attachments: ODT PAT Extension Matrix.xlsx

Dear Sanjeev and PAT Leads,

Pls find the attached file with extension matrix and engine calibration details for Truck Models.

Regards
Manoranjan

From: MANORANJAN SAHU [TMCV, Engineering, Jamshedpur]
Sent: 16 May 2025 11:57
To: SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; PANKAJ HAWARE [TMCV, Engineering, Pune] <pankaj.haware@tatamotors.com>
Cc: PAVAN SINDGIKAR [TMCV, Engineering, Pune] <pavan.s@tatamotors.com>; VISHNU DIXIT [TMCV, Engineering, Jamshedpur] <vishnud@tatamotors.com>; RAVINDRA DESHMUKH [TMCV, Engineering, Pune] <ravindra.deshmukh@tatamotors.com>; PRANAB SOU MANDAL [TMCV, Engineering, Jamshedpur] <pranab.soumandal@tatamotors.com>; DEBABRATA DAS [TMCV, Engineering, Jamshedpur] <DDD326082@tatamotors.com>; SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; PALASH DAS [TMCV, Engineering, Jamshedpur] <palash.das@tatamotors.com>; RAVI RANJAN [TMCV, Engineering, Jamshedpur] <ranjan.ravi@tatamotors.com>; SUJEET PRASAD [TTL, ER&D TML_iSource_CVBU] <sujeetp.ttl@tatamotors.com>; SUBHANKAR BANERJEE [TTL, ER&D TML_iSource_CVBU] <sb.ttl@tatamotors.com>; RANA SINGH [TMCV, Engineering, Jamshedpur] <Rana.Singh1@tatamotors.com>; JITENDRA KUMAR SINGH [TMCV, Engineering, Jamshedpur] <jitendrakumar.singh@tatamotors.com>; AMIT KHOT [TTL, ER&D TML_iSource_CVBU] <ak926960.ttl@tatamotors.com>; SANTOSH KUMAR [TMCV, Engineering, Pune] <santosh.kumar3@tatamotors.com>; NAROTTAM PANKAJ [TMCV, Engineering, Pune] <narottam.pankaj@tatamotors.com>
Subject: RE: HCV Truck_FE/PD Beta Validation FIR/Test Report- All lead & follower models name to be mention

Dear Sanjeev and PAT team leads,

This is with ref to the ESO team requirements for AC regulation and recent developments of 320 HP engines across the platform, request you to ensure **all lead & follower models name to be mentioned in PAT FIR/Test report accordingly** as per the attached variant matrix for Trucks.

Similar sheet will be shared by the other platform teams members.

CC: Rana, Jitendra , Santosh, and Narottam ji : Pls add your variant models on the attached matrix and share urgently.

Regards
Manoranjan

From: PRASHANT FATING [TMCV, Engineering, Pune] <prashant.fating@tatamotors.com>
Sent: 26 February 2025 13:36
To: MANORANJAN SAHU [TMCV, Engineering, Jamshedpur] <manoranjan.sahu@tatamotors.com>; NAROTTAM PANKAJ [TMCV, Engineering, Pune] <narottam.pankaj@tatamotors.com>; NIKHIL MATHUR [TMCV, Engineering, Pune] <NMM326121@tatamotors.com>; PARAG PHIRKE [TMCV, Engineering, Pune] <parag.phirke@tatamotors.com>; SUBHANKAR BANERJEE [TTL, ER&D TML_iSource_CVBU] <sb.ttl@tatamotors.com>; AMIT KHOT [TTL, ER&D TML_iSource_CVBU] <ak926960.ttl@tatamotors.com>; KUNDAN KUMAR [TMCV, Engineering, Jamshedpur] <kundan.kumar1@tatamotors.com>; PRITHVIRAJ RAY CHAUDHURI [TMCV, Engineering, Jamshedpur] <prithviraj.rc@tatamotors.com>; BIMAL KISHOR RAJAK [TTL, ER&D TML_iSource_CVBU] <bimalkis.ttl@tatamotors.com>; PRAMOD SAHU [TTL, ER&D TML_iSource_CVBU] <pramodsa.ttl@tatamotors.com>; PRAVIN KUMAR [TTL, ER&D TML_iSource_CVBU] <pravin.ttl@tatamotors.com>; DHIRAJ SINGH [TTL, ER&D TML_iSource_CVBU] <d0001001.ttl@tatamotors.com>
Cc: BHUSHAN KOTHARI [TMCV, Quality, Pune] <bkothari@tatamotors.com>; RAVINDRA DESHMUKH [TMCV, Engineering, Pune] <ravindra.deshmukh@tatamotors.com>; ASHISH DIGIKAR [TMCV, Quality, Pune] <ashish.digikar@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; SHASHWAT VISHWAKARMA [TTL, Engg Quality, ER&D TML_PO Onsite_EQ] <sv925274.ttl@tatamotors.com>; ROHIT SINKAR [TTL, Engg Quality, ER&D TML_PO Onsite_EQ] <rs926394.ttl@tatamotors.com>; SHUBHAM JAIN [TTL, ER&D TML_iSource_CVBU] <shubhamj.ttl@tatamotors.com>
Subject: HCV Truck_FE/PD Beta Validation FIR/Test Report- All lead & follower models name to be mention

Hello HCV Truck Team,

As per approved PAT extension matrix, **FE/PD testing (Beta Validation) to be done on -4830.T (6.7L), 3523.T (5.6) & 4221.T (5L),**

Request to you, Coordinate with ODT team for ...All lead & follower models name to be mention in PAT FIR/Test Report accordingly.

Reference list (6.7L, 5.6L - received from your side) attached -Lead/Follower model name

SIGNA AC PAT Extension Matrix (HCV- Truck, Tipper and Tractor)				Date: 25.09.2024 , REV - NR
	6.7 Ltr (Cummins) - Diesel	5.6 Ltr (Cummins) - Diesel	5Ltr (TML)- Diesel	Remarks
A1: Performance & Drivability	TBD: 5530.S / 4830.T / 4830.TK / 4225.T (G950) / SIGNA 2830.K REPTO	TBD: 4623.S / 3523.T / 4018.S	TBD: 5521.S / 4221.T / 2821.TK (RMC)	Based on the testing as per WCC, other tonnage node of 6.7 Ltr , 5.6 Ltr and 5 Ltr will get extended for all the platform
A2: Fuel Economy	TBD: 5530.S / 4830.T / 4830.TK/ SIGNA 2830.K REPTO	TBD: 4623.S / 3523.T / 2818.TK (RMC)	TBD: 5521.S / 4221.T / 2821.TK (RMC)	Based on the testing as per WCC, other tonnage node of 6.7 Ltr , 5.6 Ltr and 5 Ltr will get extended for all the platform

COWL AC PAT Extension Matrix (Truck)				Date: 25.09.2024 , REV - NR
	6.7 Ltr (Cummins) - Diesel	5.6 Ltr (Cummins) - Diesel	5Ltr (TML)- Diesel	Remarks
A1: Performance & Drivability	C/O From Base SIGNA	C/O From Base SIGNA	C/O From Base SIGNA	Based on the testing as per WCC, other tonnage node of 6.7 Ltr , 5.6 Ltr and 5 Ltr will get extended for all the platform
A2: Fuel Economy	C/O From Base SIGNA	C/O From Base SIGNA	C/O From Base SIGNA	Based on the testing as per WCC, other tonnage node of 6.7 Ltr , 5.6 Ltr and 5 Ltr will get extended for all the platform

Regards,
Prashant Fating
ESO-Engineering Signoff

ODT FIR Extension Matrix					
Sl No	Vehicle PAT	Engine	Lead vehicle / Vehicle Tested	Truck Models	Engine Cal Id
1	FE	6.7L (300HP)	SIGNA/LPT 4830.T	SIGNA/LPT 4830.T	PE10190
2				SIGNA/LPT 4930.T	PE10190
3				SIGNA/LPT 4832.T	PE10210
4				SIGNA/LPT 4932.T	PE10210
5				SIGNA/LPT 4425.T	PE10192
6				SIGNA/LPT 4225.T	PE10192
7				SIGNA/LPT 3725.T	PE10192
8				SIGNA/LPT 3525.T	PE10192
9				SIGNA/LPT 3125.T	PE10192
10		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA/LPT 3723.T	PF10040
11				SIGNA/LPT 3523.T	PF10040
12				SIGNA/LPT 3123.T	PF10040
13				SIGNA/LPT 3023.T	PF10040
14				SIGNA/LPT 2823.T	PF10040/PF10046
15				SIGNA/LPT 2623.T	PF10040
16		5L (210HP)	SIGNA/LPT 3521.T	SIGNA/LPT 3721.T	
17				SIGNA/LPT 3521.T	
18				SIGNA/LPT 3121.T	
19				SIGNA/LPT 3021.T	
20				SIGNA/LPT 2821.T	
21				SIGNA/LPT 2621.T	

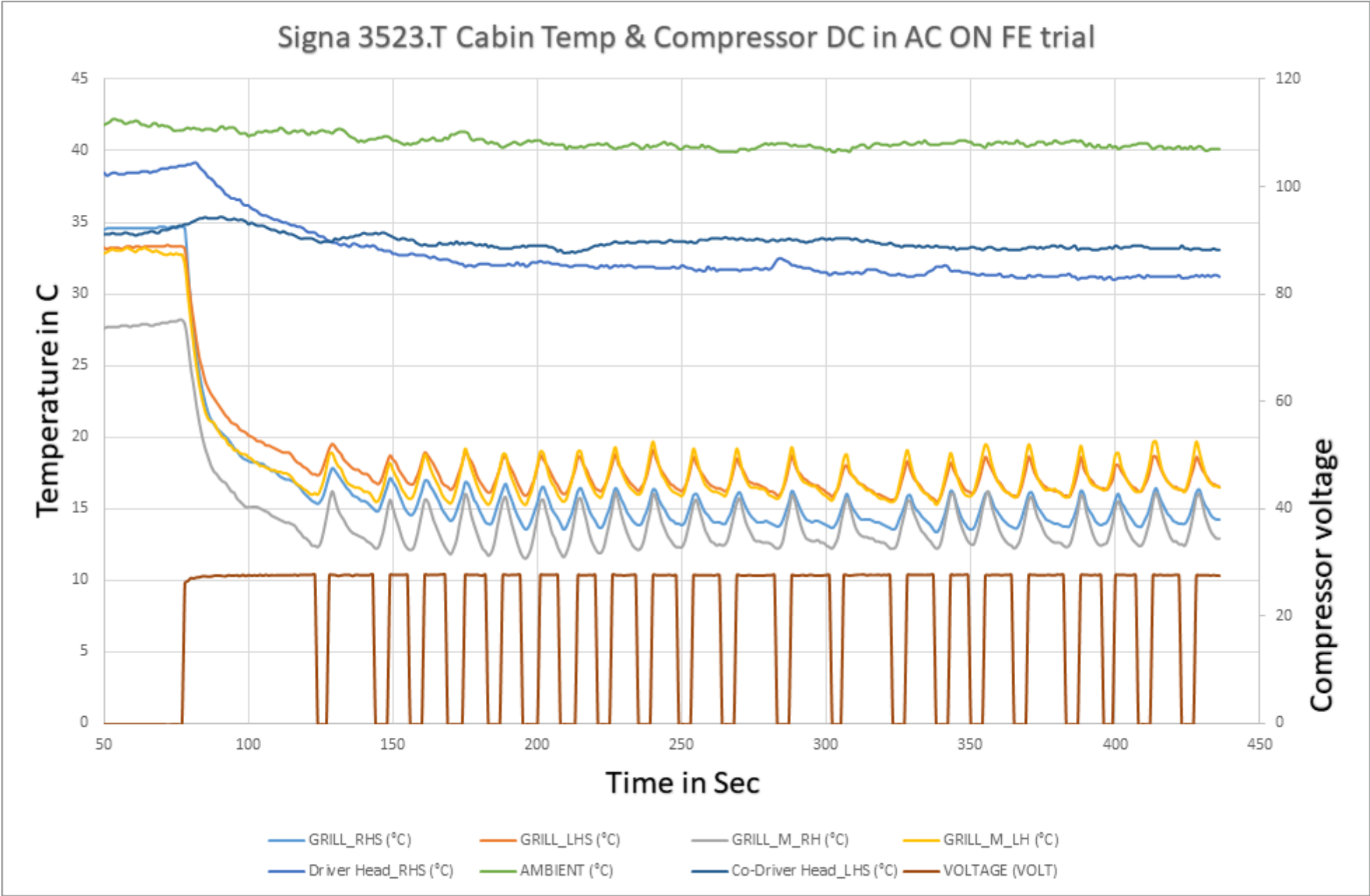
From: PALASH DAS [TMCV, Engineering, Jamshedpur]
Sent: 15 May 2025 14:15
To: MANORANJAN SAHU [TMCV, Engineering, Jamshedpur]
Cc: PAVAN SINDGIKAR [TMCV, Engineering, Pune]; AJAYKUMAR JADHAV [TMCV, Engineering, Pune]; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur]; SWAPNIL JADHAV [TMCV, Engineering, Pune]; SUBHAYU DAS [TMCV, Engineering, Jamshedpur]; DEBABRATA DAS [TMCV, Engineering, Jamshedpur]; KUNDAN KUMAR [TMCV, Engineering, Jamshedpur]
Subject: Signa 3523.T AC FE Trial – Cabin Temperature & Compressor Duty Cycle Analysis

Dear Sir,

Please find below the analysis of the Signa 3523.T AC FE trial with respect to Cabin Temperature and Compressor DC performance:

- During the trial, the AC system was operated in Eco mode with the blower at 2nd speed, and the vehicle was driven steadily at 40 kmph.
- The compressor was observed to engage and disengage in a cyclic manner.

Note: Average Compressor Engagement % time throughout the duty cycle is 65-70% (refer the below graph to for Cab temp & Compressor duty cycle)



This is for your information & further reference please.

Regards
Palash Das

Annexure 6

GURPRIT SINGH [TMCV, Engineering, Pune]

From: GURPRIT SINGH [TMCV, Engineering, Pune]
Sent: 21 May 2025 17:16
To: RANA SINGH [TMCV, Engineering, Jamshedpur]; AJAYKUMAR JADHAV [TMCV, Engineering, Pune]
Cc: SWAPNIL JADHAV [TMCV, Engineering, Pune]; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur]; Bokam Surya SashiKanth; Sweta Kumari; Ashwani Kumar Gupta; Raj Mohanty; Anil S Patil; Anish Kumar; DHEERAJ KUMAR [TMCV, Engineering, Jamshedpur]
Subject: RE: Calibration no for tipper extension matrix

Dear Rana Ji
As per attached excel, we are considering below Cal ids against extension matrix.

ODT FIR Extension Matrix					
				Tipper Models	
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	CAL ID
1	FE	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	PE10197.01
2				SIGNA 4230.TK	PE10197.01
3				SIGNA 3530.K/TK	PE10185.00
4				SIGNA 2830.TK/TK	PE10185.00
5				SIGNA 4832.TK	PE10212.00
6				SIGNA 4232.TK	PE10212.00
7				SIGNA 3532.TK/TK	PE10211.01
8				SIGNA 2832.K/TK	PE10211.01
9				SIGNA 2825.K/TK	PE10188.00
10		6.7L (300HP)	SIGNA 2830.TK REPTO	SIGNA 2830.TK REPTO	PE10185.00
				SIGNA 2832.K REPTO	PE10211.01
11		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	PE10050.00
13		5.6L (180HP)	SIGNA 2818.K	SIGNA 2818.K	PE10051.00
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	
15		5.7L CNG	Signa 2820.K	Signa 2820.K	

ODT FIR Extension Matrix					
				Tipper Models	
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	CAL ID
1	PD	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	PE10197.01
2				SIGNA 4230.TK	PE10197.01
3				SIGNA 3530.K/TK	PE10185.00
4				SIGNA 2830.K/TK	PE10185.00
				SIGNA 2830.K REPTO	PE10185.00
5				SIGNA 4832.TK	PE10212.00
6				SIGNA 4232.TK	PE10212.00
7				SIGNA 3532.K/TK	PE10211.01
8				SIGNA 2832.K/TK	PE10211.01
9				SIGNA 2832.K REPTO	PE10211.01
10		6.7L (25HP)	SIGNA/LPT 4225.T	SIGNA 2825.K/TK	PE10188.00
11		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	PE10050.00
13		5.6L (180HP)	SIGNA 4018.S	SIGNA 2818.K	PE10051.00
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	
15		5.7L CNG	Signa 2820.T	Signa 2820.K	

Regards
Gurprit Singh

From: RANA SINGH [TMCV, Engineering, Jamshedpur] <Rana.Singh1@tatamotors.com>
Sent: 21 May 2025 17:05
To: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; GURPRIT SINGH [TMCV, Engineering, Pune] <Gurprit.Singh@tatamotors.com>
Cc: SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; Bokam Surya SashiKanth <bokam.surya.sashikanth@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Ashwani Kumar Gupta <ashwani.kumar.gupta@cummins.com>; Raj Mohanty <raj.mohanty@cummins.com>; Anil S Patil <anil.s.patil@cummins.com>; Anish Kumar <kumar.anish@cummins.com>; DHEERAJ KUMAR [TMCV, Engineering, Jamshedpur] <kumar.dheeraj@tatamotors.com>
Subject: RE: Calibration no for tipper extension matrix
Importance: High

Hi Ajay/Gurprit

Refer attached matrix for with CAL ID.

With Regards

Rana singh

From: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>
Sent: 21 May 2025 16:08
To: RANA SINGH [TMCV, Engineering, Jamshedpur] <Rana.Singh1@tatamotors.com>
Cc: SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Ashwani Kumar Gupta <ashwani.kumar.gupta@cummins.com>; Raj Mohanty <raj.mohanty@cummins.com>; Anil S Patil <anil.s.patil@cummins.com>; Anish Kumar <kumar.anish@cummins.com>; GURPRIT SINGH [TMCV, Engineering, Pune] <Gurprit.Singh@tatamotors.com>
Subject: RE: Calibration no for tipper extension matrix

Dear Rana,
Kindly check and verify as we are going to updated report based on your feedback.

Regards,
Ajay

From: Anish Kumar <kumar.anish@cummins.com>
Sent: 21 May 2025 16:07
To: GURPRIT SINGH [TMCV, Engineering, Pune] <Gurprit.Singh@tatamotors.com>; RANA SINGH [TMCV, Engineering, Jamshedpur] <Rana.Singh1@tatamotors.com>
Cc: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Ashwani Kumar Gupta <ashwani.kumar.gupta@cummins.com>; Raj Mohanty <raj.mohanty@cummins.com>; Anil S Patil <anil.s.patil@cummins.com>
Subject: RE: Calibration no for tipper extension matrix

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Hello Team,

Please refer below calibration IDs for vehicles models, Kindly note that we have added one column for engine configuration based on Rana’s input and added calibration code against that.

Calibration code changes if the engine configuration changes.

ODT FIR Extension Matrix						
				Tipper Models		
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant		Calibration ID
1	FE	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	VI+Front EMP+3”	PE10197
2				SIGNA 4230.TK	VI+front EMP+3”	PE10197
3				SIGNA 3530.K/TK	Coal+Front EMP+4”	PE10185
4				SIGNA 2830.K/TK	Coal+Front EMP+4”	PE10185
5				SIGNA 4832.TK	VI+Front EMP+3”	PE10213
6				SIGNA 4232.TK	VI+front EMP+3”	PE10213
7				SIGNA 3532.K/TK	Coal+Front EMP + 4”	PE10234
8				SIGNA 2832.K/TK	Coal+Front EMP + 4”	PE10234
9				SIGNA 2825.K/TK	VI + Tri-EGTS+front EMP	PE10188
10		6.7L (300HP)	SIGNA 2830.TK REPTO	SIGNA 2832.K REPTO		PE10185 (same as normal tipper)
11		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	Coalescer breather+Tri EGTS+Rear Emp	PF10050
13		5.6L (180HP)	SIGNA 2818.K	SIGNA 2818.K	Coal+Tri-EGTS	PF10051 (RMC)
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	TML	NA
15		5.7L CNG	Signa 2820.K	Signa 2820.K	TML	NA

Best Regards
Anish Kumar
Technical Specialist – FE Cell
Cell: 9264433836

From: Gurprit.Singh@tatamotors.com <Gurprit.Singh@tatamotors.com>
Sent: 21 May 2025 15:10
To: Rana.Singh1@tatamotors.com
Cc: ajaykumar_jadhav@tatamotors.com; Swapnil.Jadhav@tatamotors.com; sanjeev1.kumar@tatamotors.com; Anish Kumar <kumar.anish@cummins.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>
Subject: Calibration no for tipper extension matrix

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Dear Ranaji

We discussed, please provide calibration no for Tipper extension matrix.

ODT FIR Extension Matrix					
				Tipper Models	
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	Calibration ID
1	FE	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	
2				SIGNA 4230.TK	
3				SIGNA 3530.K/TK	
4				SIGNA 2830.K/TK	
5				SIGNA 4832.TK	
6				SIGNA 4232.TK	
7				SIGNA 3532.K/TK	
8				SIGNA 2832.K/TK	
9				SIGNA 2825.K/TK	
10		6.7L (300HP)	SIGNA 2830.TK REPTO	SIGNA 2832.K REPTO	
11		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	
13		5.6L (180HP)	SIGNA 2818.K	SIGNA 2818.K	
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	
15		5.7L CNG	Signa 2820.K	Signa 2820.K	

ODT FIR Extension Matrix					
				Tipper Models	
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	Calibration ID
1	PD	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	
2				SIGNA 4230.TK	
3				SIGNA 3530.K/TK	
4				SIGNA 2830.K/TK	
5				SIGNA 2830.K REPTO	
6				SIGNA 4832.TK	
7				SIGNA 4232.TK	
8				SIGNA 3532.K/TK	
9				SIGNA 2832.K/TK	
10				SIGNA 2832.K REPTO	
11		6.7L (25HP)	SIGNA/LPT 4225.T	SIGNA 2825.K/TK	
12		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	

13		5.6L (180HP)	SIGNA 4018.S	SIGNA 2818.K	
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	
15		5.7L CNG	Signa 2820.T	Signa 2820.K	

Regards
Gurprit Singh

From: GURPRIT SINGH [TMCV, Engineering, Pune]
Sent: 21 May 2025 12:02
To: Anish Kumar <kumar.anish@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>
Cc: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>
Subject: RE: HCV_AC project PED validation status

Dear Anish
Request you to also provide the details of the following models

Model	ENG	ERC No	Calibration ID	CAL Used in Vehicle
SIGNA 4230.TK				
SIGNA 3530.K/TK				
SIGNA 2830.K/TK				
SIGNA 4832.TK				
SIGNA 4232.TK				
SIGNA 3532.K/TK				
SIGNA 2832.K/TK				
SIGNA 2825.K/TK				
Signa 2823 RMC				

Regards
Gurprit Singh

From: Anish Kumar <kumar.anish@cummins.com>
Sent: 20 May 2025 15:22
To: SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; Sweta Kumari <sweta.kumari@cummins.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; HCV PED <MHCVPED@tatamotors.onmicrosoft.com>; GURPRIT SINGH [TMCV, Engineering, Pune] <Gurprit.Singh@tatamotors.com>
Subject: RE: HCV_AC project PED validation status

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Hello Ajay/Swapnil,

Please refer below CAL-ID for priority tested vehicles.

Sr. No.	Model	ENG	ERC Number	Calibration ID	CAL Used in Vehicle
1	Signa 5530.S 6x4	6.7L 300HP	ERC1283	PE10143.03	XYPAA17D04
2	Cowl 4830.T		EJ00913	PE10182.01	PE10182.01
3	Signa 4830.T		ERC1210	PE10190.02	XYRAA17E85
4	Signa 4830.TK		ERC1261	PE10003.15	XYPAA17D91

5	Signa 2830.K REPTO		ERC1237	PE10206.02	XYPAA17E07
6	Signa 4225.T (G950)	250HP	ERC1282	XYPAA17D17	XYPAA17F01
7	Signa 3523.T	5.6L 230 HP	ERC1309	PF10040.02	XWTAA19D07
8	Signa 4623.S		EJ00935	PF10002.09	XWTAA19D14
9	Signa 4023.S - 750NM		EJ00931	PF10070	XWTAA19D10
10	Signa 2818.TK RMC	5.6L 180HP - 700Nm	EJ00936	PF10019.02	XWTAA13D15

Best Regards
Anish Kumar
Technical Specialist – FE Cell
Cell: 9264433836

From: Swapnil.Jadhav@tatamotors.com <Swapnil.Jadhav@tatamotors.com>
Sent: 20 May 2025 12:30
To: Anish Kumar <kumar.anish@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Bokam Surya SashiKanth <bokam.surya.sashikanth@cummins.com>
Cc: ajaykumar_jadhav@tatamotors.com; MHCVPED@tatamotors.onmicrosoft.com
Subject: RE: HCV_AC project PED validation status

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Dear Surya,

Instead of revision number please mention CVN number

Best Regards,
Swapnil Jadhav
PED - HCV
ERC Pune - Vehicle Engg. (CVBU)
Mo. No. - +91 9922001494



From: SWAPNIL JADHAV [TMCV, Engineering, Pune]
Sent: 20 May 2025 11:26
To: Anish Kumar <kumar.anish@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>
Cc: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; HCV PED <MHCVPED@tatamotors.onmicrosoft.com>
Subject: FW: HCV_AC project PED validation status

Dear Anish,

As discussed, PI revert below table with Calibration ID & revision number.

Its Urgent.

Sr. No.	Model	ENG	Calibration ID	CVN
---------	-------	-----	----------------	-----

1	Signa 5530.S 4X2	6.7L 300HP		
2	Cowl 4830.T			
3	Signa 4830.T			
4	Signa 4830.TK			
5	Signa 2830.K REPTO			
6	Signa 4225.T (G950)	250HP		
7	Signa 3523.T	5.6L 230 HP		
8	Signa 4623.S			
9	Signa 4023.S 925NM			
10	Signa 2818.TK RMC	5.6L 180HP		
11	Signa 5521.S	5L 210HP		
12	Signa 3521.T (5.57)			
13	Signa 2821.TK RMC			
14	LPK 2821.TK 709 Cab RMC			
15	Signa 2820.T CNG	5.7L CNG		
16	Cowl 2820.T CNG			

Best Regards,
Swapnil Jadhav
PED - HCV
ERC Pune - Vehicle Engg. (CVBU)
Mo. No. - +91 9922001494



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